

Gauge Architecture and Gauge Skin in the Standard Model

Integrated Role-Occupancy Closure and Locus-Sensitive Gauge Audit

Central result. The primary theorem is proved on one integrated SYM-1 gauge target rather than on a collection of closure-friendly microtargets. Eleven occupant-independent target roles are fixed before residue inspection. A target-derived protocol generates the candidate-locus universe, including atlas-external objects and genuine presentation duplicates. All residue/role pairs are classified by certified capability criteria; all 4,608 pointwise eligible injective preassignments are evaluated against independently computed transport, quotient, representation, anomaly, boundary, topology, theta, BRST, and synchronization constraints. One compatible complete assignment remains, and its lawful assignment quotient has one orbit. A separate two-chart target exhibits a nontrivial automorphism group and two assignments in one lawful orbit. Proper compact-support gauge operations remain quotient-trivial skin at the proper-bulk locus without demoting the presentation, comparison, reduction, response, and consistency architecture.

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Abstract

This paper separates the load-bearing gauge architecture of the Standard Model from representative gauge operations without treating AASC as an optional interpretation of independently complete physics. Determinate fields, states, transformations, quotients, records, and admissible continuations already instantiate the Admissibility–Standing–Reference–Irreversibility kernel. Conventional gauge theory supplies the target-specific transformation, quotient, boundary, topology, representation, cohomology, and anomaly facts used by the classification.

The closure engine is Role-Occupancy Closure under Admissibility. Unlike earlier versions, the primary closure theorem is evaluated on one integrated Standard Model gauge target containing coherent presentation, comparison and transport, physical reduction, matter response, quantum consistency, proper-bulk operations, boundary/asymptotic action, global-form/topology data, large-transformation/ θ realization, BRST/BV realization, and cross-realization synchronization. The target role set is fixed before candidate inspection. The candidate universe is generated from a frozen physical target signature and includes atlas-backed views, target-generated composite packages, atlas-external diagnostics, and multiple nontrivial presentation-equivalence stress pairs. Every residue-role pair receives a positive or negative certificate. Pointwise eligibility is not treated as realization: every injective eligible preassignment is evaluated by a named compatibility predicate whose components are computed from the candidate profiles. Live residue, complete assignments, lawful automorphisms, and assignment orbits are then derived. The integrated target has one compatible assignment and one lawful orbit. A controlled two-chart target demonstrates that nontrivial lawful automorphisms can identify multiple assignments without erasing physical distinctions.

Target-role occupancy, ATS anchor/tensor/skin type, and SYM-0 status remain independent classification axes. Witnesses, anomaly contributions, charge values, boundary-status tables, obstruction records, and downstream handoffs are not promoted into architectural roles. The endpoint is complete relative to the frozen integrated SYM-1 target signature and certificate; any same-target missing role, candidate, eligibility edge, compatibility condition, or lawful-equivalence invariant is an explicit falsifier that reopens the theorem. Full Standard Model symmetry-atlas exhaustion remains assigned to SYM-11.

Read This First

The theorem spine

kernel necessity $\implies$ integrated fixed target and certificate
 $\implies$ occupant-independent target-role set
 $\implies$ target-generated candidate-locus residue
 $\implies$ complete certified eligibility matrix
 $\implies$ all eligible preassignments and computed compatibility
 $\implies$ derived live residue and complete assignments
 $\implies$ full-invariant lawful automorphisms and one orbit
 $\implies$ independent ATS and SYM-0 typing.

One integrated theorem target. The proper-bulk, boundary, theta, anomaly, global-form, and BRST scopes remain registered as projection and evidence targets. They are not used as separate closure-friendly proof universes. The principal occupancy theorem runs over their integrated parent target.

Kernel-first in necessity; QFT-first in record certification. The explanatory order begins with the necessity conditions of determinate objecthood. The evidentiary order for an individual gauge record begins with ordinary gauge theory: target, transformation, quotient, observables, boundary, topology, representation, quantum regime, and anomaly or cohomology facts are fixed before eligibility, compatibility, ATS type, or SYM-0 status is certified.

What is unique. The theorem establishes one compatible occupation orbit of the integrated target architecture relative to a role-realization-complete finite certificate. It does not assert that every mathematical presentation or explanatory narrative is literally identical. The lawful quotient removes only transformations preserving the full target, role-incidence, eligibility, compatibility, standing, tensor, quotient, boundary, topology, continuation, anomaly, and failure tuple.

Exact falsification routes. The result is reopened by a missing same-target role, a target-relevant object absent from the generator, an overbroad or incomplete presentation quotient, a false positive or false negative eligibility edge, an omitted compatibility component, a live residue absent from assignment analysis, a second non-equivalent assignment orbit, or an overbroad automorphism quotient. These are theorem-breaking routes, not matters of preference.

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Dependency Order, Kernel Activation, and Role-Occupancy Closure

1.1 AASC is not an interpretive overlay

AASC identifies the minimal necessity conditions of non-degenerate determinate objecthood. A physical gauge domain containing determinate carriers, states, transformations, quotients, records, or endpoints therefore instantiates Admissibility, Standing, Reference, and Irreversibility. The kernel is not inferred from gauge-theory notation, and its authority does not depend on a preferred gauge ontology. The fixed-domain necessity and anti-weakening results are imported from the kernel corpus [14, 20].

1.2 Unique gate and admissible interior

The fixed-domain and apex theorems exclude competing same-domain gates, hidden selectors, retrospective repair, and discrete plural admissible interiors [14, 27]. They fix the admissibility environment in which target roles, candidate residue, eligibility, compatibility, and assignment are assessed. They do not by themselves prove a unique downstream factorization of gauge architecture.

Theorem 1.1 (Imported fixed-domain gate discipline). Within one fixed standing-instantiated domain, a second standing-relevant gate, boundary selector, representative-sensitive authorizer, or retrospective repair is not a same-scope refinement. A genuine alteration changes the certificate or target; a difference changing no standing-bearing consequence is downstream presentation.

Claim status: Imported from the kernel, fixed-domain, and apex closure papers.

1.3 Role occupancy is the downstream closure engine

For a fixed certificate C and target T , let $R_T(C, T)$ be the target-forced role set, $\text{Residue}_T^{\text{live}}(C, T)$ the live target-relative residue, E_C the eligibility relation, J_C the joint-compatibility predicate, and $\Gamma_{C,T}^{\text{lawful}}$ the automorphism group preserving the full certificate invariant tuple. The compatible complete assignments are

$$\Sigma_C(T) = \left\{ \sigma : R_T(C, T) \xrightarrow{\sim} \text{Residue}_T^{\text{live}}(C, T) : E_C(\sigma(r), r) \ \forall r, \ J_C(\sigma) = 1 \right\}.$$

For a role-realization-complete certificate,

$$\text{RealizationLicensed}(C, T) \iff \left| \Sigma_C(T) / \Gamma_{C,T}^{\text{lawful}} \right| = 1.$$

This theorem is imported from the dedicated role-occupancy and role-cardinality sources [23, 24] and specialized here to the integrated Standard Model gauge target.

Theorem 1.2 (Role-Occupancy Closure specialization). If the integrated SYM-1 certificate satisfies candidate, quotient, role-set, eligibility, compatibility, live-residue, and failure completeness, realization is licensed exactly when its complete compatible assignments have one orbit under certificate-lawful automorphisms preserving the full invariant tuple.

Claim status: Imported general theorem plus finite target-specific specialization.

1.4 Strict division of evidentiary labor

The dependency stack is non-circular.

1. The kernel fixes the necessity conditions of determinate standing-bearing use.
2. The integrated target declaration fixes the genuine architecture and operation roles before candidates are inspected.
3. Conventional QFT fixes transformation, quotient, representation, charge, boundary, topology, anomaly, and cohomology facts.
4. Those facts determine positive and negative eligibility edges through role criteria.
5. Co-instantiation constraints determine compatibility and live residue.
6. Exhaustive finite enumeration computes complete assignments and lawful orbits.
7. ATS and SYM-0 type the resulting descriptive elements and statuses on independent axes.

No assignment determines the role set, candidate generator, presentation quotient, eligibility criteria, compatibility predicate, or lawful automorphism group.

1.5 Projection targets are evidence scopes, not separate closure universes

The registry retains proper-bulk, boundary, topology, theta, BRST, anomaly, global-form, governance, Gribov, and electroweak-handoff scopes. Their purpose is to state exact QFT hypotheses and loci. The principal closure theorem runs over the integrated parent target. A second controlled target with two interchangeable chart positions is retained solely to verify that a nontrivial lawful automorphism group can reduce two assignments to one orbit.

Integrated Gauge Target, Projection Scopes, and Quotient Discipline

2.1 The primary integrated target

The primary theorem target is $D_{\text{SM}}^{\text{int}}$, the registered object D-SYM1-SM-ASSEMBLY. It contains simultaneously:

- local presentation and overlap data;
- paths, connections, transport, curvature, and holonomy;
- constraints, generators, the classical physical quotient, and the governance-to-physical bridge;
- the Standard Model chiral representation and charge ledgers;
- aggregate local and global gauge-consistency conditions;
- proper compact-support operations;
- boundary/asymptotic action, charge, flux, integrability, and edge-data structure;
- an admitted G_{F} family, fixed topology data, large transformations, and theta-sector representation;
- a declared perturbative BRST/BV realization with separate state, local, and relative cohomological interfaces;
- cross-realization synchronization.

The target is sharp-local in the ARC sense: full line-operator and generalized-symmetry completion remains SYM-9 material.

2.2 Projection scopes

The older registered scopes remain available as restrictions of the integrated target and as evidence ledgers. They isolate the assumptions of the proper-bulk quotient theorem, boundary charge analysis, theta classification, BRST cohomologies, anomaly tests, global-form preview, Gribov obstruction, governance quotient, and electroweak handoff. They are not separately counted as independent closure results.

Theorem 2.1 (Integrated-target fixation). Every SYM-1 gauge claim is either a well-typed candidate-locus record of the integrated target, a registered projection of that target, an explicit target update, or a same-target protocol gap. A change of global form, boundary, topology, field content, operator category, quantum realization, vacuum, or scale is a target update unless already parameterized by the integrated target.

2.3 Typed quotient family

The quotient discipline distinguishes

$$\begin{aligned}
 Q_D^{\text{gov}} &= P_D / \sim_{\text{future}}, & Q_D^{\text{cl}} &= \mathcal{C}_D / \mathcal{G}_{0,D}, \\
 Q_D^{\text{red}} &= \mathcal{C}_D^{\text{constraint}} / \ker \omega_C, & Q_D^{\text{ray}} &= (\mathcal{H}_D \setminus \{0\}) / \mathbb{C}^\times, \\
 Q_D^{\text{BRST, state}} &= H^0(Q_{\text{BRST}}, \mathcal{H}_{\text{ext}}), & Q_D^{\text{BRST, local}} &= H^0(s, \mathcal{A}_{\text{loc}}), \\
 Q_D^{\text{BRST, rel}} &= H^{g,n}(s \mid d).
 \end{aligned}$$

No two are identified merely because both remove a use of the word “gauge.” Every comparison has a declared domain, codomain, hypothesis bundle, and failure mode.

Theorem 2.2 (Proper-bulk quotient-fidelity bridge). Under gauge invariance of the admitted tests, separation of proper-gauge orbits, fixed boundary and topology, and the stated regularity hypotheses, the governance and classical proper-gauge equivalence relations coincide on the declared locus. If separation or regularity fails, the bridge weakens to a comparison map or remains unresolved.

Claim status: Conditional local theorem; the physical proper-gauge theorem can also be stated directly on Q_D^{cl} .

2.4 Anomaly as failure of descent

The anomalous case is not represented by assuming a successful quotient. The measure, effective action, or BRST structure either descends under the declared assumptions or yields a local/global obstruction certificate. The integrated consistency role is occupied only by the aggregate anomaly-neutral package; individual coefficients and cancellation witnesses remain status and evidence records.

Quotient-Fidelity Bridge Registry

Bridge	Source	Target	Relation	Assumptions	Failure mode
QB-01-GOV-C L	Q-GOV-STANDI NG	Q-CL-PROPER- CONFIG	Conditiona lIsomorphi sm	admitted tests and con- tinuations are gauge in- variant; they separate proper-gauge reduced con- figurations; boundary and topology are fixed; proper action is regular enough for the orbit quotient	If tests fail to sepa- rate or boundary/- topology changes, only a comparison map or no bridge is licensed.

QB-02-CL-PH ASE	Q-CL-PROPER- CONFIG	Q-REDUCED-PH ASE	Conditiona lCompatibi lity	regular constraint sur- face; presymplectic ker- nel equals proper gauge distribution; observable separation and maximality	Singular strata or boundary generators require a refined target.
QB-03-DIRAC -BRST-STATE	Q-REDUCED-PH ASE	Q-BRST-STATE	Conditiona lCompariso n	nilpotent anomaly-free BRST charge; appropriate state complex and inner product; no unaccounted global gauge obstruction	Failure of nilpotency or state-space hy- potheses blocks the bridge.
QB-04-CL-BR ST-LOCAL	Q-CL-PROPER- CONFIG	Q-BRST-LOCAL	Conditiona lCompariso n	regular irreducible gauge theory in the selected lo- cal complex; nilpotent BRST differential; appro- priate on-shell/off-shell convention fixed	Open/reducible algebras or changed complex require a target update.
QB-05-REL-A NOMALY	Q-BRST-RELAT IVE	Q-QUANTUM-DE SCENT	Diagnostic NotIsomorp hism	local perturbative anomaly analysis; regu- larized quantum effective action; descent equations in the selected complex	Global anomalies and nonperturbative sectors are separate records.
QB-06-RAY-T HETA	Q-PROJECTIVE -RAY	Q-LARGE-DECL ARED	PhaseInsuf ficiencyRu le	physical pure states are rays; large-transformation phase acts uniformly on one state vector	A separate sector/- operator/interfer- ence certificate can establish nontrivial action.

Three-Axis Typing, Candidate Residue, and Certificate Discipline

3.1 Three independent questions

Every gauge record carries three logically separate classifications.

1. **Target-role occupancy:** which target-forced architectural or operation position, if any, does the residue occupy?
2. **ATS element type:** does the declared variation change target definedness, quotient-level consequence, or only presentation—anchor, tensor, or skin?
3. **SYM-0 status:** what are the quotient action, physical-action, anomaly, boundary, vacuum, scale, and epistemic statuses?

A proper local gauge operation may occupy the proper-operation slot, be ATS skin at the proper-bulk quotient locus, and have quotient-trivial SYM-0 status. A gauge anomaly occupies no successful architectural role while remaining an ATS failure-bearing tensor/diagnostic and a SYM-0 obstruction. The axes answer different questions and are not inferred from one another.

3.2 Candidate-locus units

The unit presented to the occupancy engine is

$$x = (\text{object}, D, \text{locus}, \text{action or relation}, q, \text{quantum scope}).$$

Thus a transformation in the proper bulk, the same formula with a nonvanishing boundary parameter, its differentiable generator, its charge value, and its Ward identity are distinct candidates. The integrated generator traverses the target signature rather than the atlas. Atlas rows are canonical evidence-backed instances of the generated families.

3.3 Presentation quotient only

Candidates are quotiented only by presentation equivalence $\sim_{\text{pres}}$. Two candidates are equivalent when target, locus, quotient behavior, tensor content, boundary/topology, representation and charge response, anomaly status, continuation, failure data, target-role capabilities, and compatibility profile agree, and the remaining difference is notation, basis, coordinate, proper-gauge representative, phase convention, or redundant bookkeeping. Role equivalence is not used to form residue [24].

Definition 3.1 (Presentation-quotiented residue). For the fixed target certificate C_D , let X_D^0 be the generated candidate-locus universe. The residue is

$$X_D = X_D^0 / \sim_{\text{pres}} .$$

The release includes several nontrivial equivalence classes: relabelled chart order, naturally equivalent transport presentation, weak-basis representation, BRST exact-representative shift, and theta phase convention.

3.4 Eligibility is role-indexed and certificate-bearing

The relation

$$E_D \subseteq X_D \times R_D$$

is evaluated only after R_D and X_D are frozen. Each role is specified by an occupant-independent target datum and a required certified capability. A positive edge is produced exactly when the residue's QFT facts include that capability. A negative edge records the missing role criterion. The desired assignment is not an input to the matrix.

Theorem 3.2 (Complete finite eligibility matrix). For the integrated target and the controlled two-chart target, every residue-role pair receives exactly one certified disposition. The integrated matrix contains substantial pointwise competition: multiple roles have more than one individually eligible residue, so assignment uniqueness is not manufactured by a one-edge incidence graph.

Claim status: Exhaustive finite certificate calculation relative to the frozen target signature.

3.5 Joint compatibility is computed independently

For an injective pointwise eligible map $\tau : R_D \hookrightarrow X_D^{\text{elig}}$, the integrated predicate evaluates

$$\begin{aligned} J_D(\tau) = & J_{\text{target}} \wedge J_{\text{presentation}} \wedge J_{\text{transport}} \wedge J_{\text{reduction}} \wedge J_{\text{response}} \\ & \wedge J_{\text{anomaly}} \wedge J_{\text{proper}} \wedge J_{\text{boundary}} \wedge J_{\Gamma, \text{top}} \wedge J_{\theta} \\ & \wedge J_{\text{BRST}} \wedge J_{\text{sync}} \wedge J_{q\text{-cross}} \wedge J_{\Gamma\theta} \wedge J_{\text{response/anomaly}} \wedge J_{\partial/\text{top}} . \end{aligned}$$

No coordinate defaults to true. Each value is calculated from the selected candidates' certified profiles. The release enumerates all 4,608 pointwise eligible integrated preassignments and records every failed component.

3.6 Source and claim status

A certificate records whether its conclusion is imported, locally proved, conditional, open, rejected, or superseded; the evidence strength; exact source locator; target assumptions; and nonclaims. The finite model checker verifies the role-occupancy conclusion over the supplied QFT facts. It does not convert an imported continuum-QFT theorem into a local proof.

The Integrated Target-Role Architecture

4.1 What counts as a target role

A target role is a load-bearing architectural or declared operation position whose absence changes the realization of the integrated target. It is not merely a status, diagnostic, charge value, anomaly coefficient, witness, obstruction, or downstream handoff. Version 9 therefore removes boundary-status tables, individual anomaly contributions, cancellation certificates, Wilson-loop witnesses, ghost-number grading, Gribov obstruction, and electroweak handoff from the target-role set. Those objects remain fully typed on the ATS and SYM-0 axes.

4.2 The eleven integrated roles

The integrated role set is fixed before residue inspection:

1. coherent local presentation architecture;
2. comparison and transport architecture;
3. physical reduction architecture;
4. matter-response architecture;
5. quantum-consistency architecture;
6. proper-bulk gauge-operation class;
7. boundary/asymptotic action architecture;
8. global-form and topology architecture;
9. large-transformation/theta realization;
10. BRST/BV realization architecture;
11. cross-realization synchronization architecture.

Role	Target datum	Necessity proposition	Cost of denial
R9-PRESENTATION	The fixed target is locally represented and requires identity, inverse, restriction, partial composition, and overlap coherence.	If the integrated SYM-1 target is realized, some certified residue must occupy R9-PRESENTATION.	Local displays cease to constitute presentations of one determinate gauge target.

Role	Target datum	Necessity proposition	Cost of denial
R9-COMPARISON	The fixed target contains carrier data at separated locations and stages requiring identity-preserving comparison.	If the integrated SYM-1 target is realized, some certified residue must occupy R9-COMPARISON.	Separated carrier data cannot be related within one gauge construction.
R9-REDUCTION	The target contains representative multiplicity that must be separated from physical multiplicity.	If the integrated SYM-1 target is realized, some certified residue must occupy R9-REDUCTION.	Gauge-related representatives cannot be distinguished from physically distinct configurations.
R9-RESPONSE	The target includes the observed chiral Standard Model representation and charge ledger.	If the integrated SYM-1 target is realized, some certified residue must occupy R9-RESPONSE.	The gauge architecture no longer acts coherently on the frozen matter carrier.
R9-CONSISTENCY	The target is a standing-bearing anomaly-free quantum gauge architecture.	If the integrated SYM-1 target is realized, some certified residue must occupy R9-CONSISTENCY.	The quantum gauge construction need not descend to a consistent physical target.
R9-PROPER-OPERATION	The target explicitly includes compact-support or boundary-identity gauge operations at the proper-bulk quotient locus.	If the integrated SYM-1 target is realized, some certified residue must occupy R9-PROPER-OPERATION.	The central architecture/operation distinction lacks its proper-operation term.
R9-BOUNDARY-ACTION	The target includes nonvanishing boundary parameters, edge data, differentiable generators, charges, and flux conditions.	If the integrated SYM-1 target is realized, some certified residue must occupy R9-BOUNDARY-ACTION.	Bulk redundancy and boundary physical action remain untyped and may be conflated.
R9-GLOBALFORM-TOPOLOGY	The target is parameterized by admitted faithful global forms and fixed bundle/topological data.	If the integrated SYM-1 target is realized, some certified residue must occupy R9-GLOBALFORM-TOPOLOGY.	Local Lie-algebra data are incorrectly treated as a complete global gauge target.
R9-THETA-REALIZATION	The target distinguishes large-transformation quotienting, wavefunctional equivariance, theta character, and certified sector action.	If the integrated SYM-1 target is realized, some certified residue must occupy R9-THETA-REALIZATION.	A common phase, sector label, and physical topological action are conflated.
R9-BRST-REALIZATION	The target includes a declared perturbative gauge-fixing/BRST realization with separated state, local, and relative cohomology interfaces.	If the integrated SYM-1 target is realized, some certified residue must occupy R9-BRST-REALIZATION.	The cohomological realization is incomplete or silently promoted to universal primitive architecture.
R9-SYNCHRONIZATION	The integrated target combines geometric, Hamiltonian, representation, anomaly, boundary, topology, and BRST modules that must share one target invariant bundle.	If the integrated SYM-1 target is realized, some certified residue must occupy R9-SYNCHRONIZATION.	Separately valid modules need not coexist as one Standard Model gauge realization.
R9-CHART-A	The controlled target contains two lawfully interchangeable chart positions.	If the two-chart target is realized, some residue must occupy R9-CHART-A.	One chart position is unoccupied.
R9-CHART-B	The controlled target contains two lawfully interchangeable chart positions.	If the two-chart target is realized, some residue must occupy R9-CHART-B.	One chart position is unoccupied.

Theorem 4.1 (Occupant-independent role necessity). For every role in the integrated role set, realization of the target requires that some certified residue occupy the role. Each necessity proof states the target datum forcing the position and the exact loss incurred by denial without naming the eventual occupant.

4.3 Role-set saturation and falsifiability

A proposed additional load-bearing burden has four dispositions:

- (i) governance-equivalent to a declared role;
- (ii) a same-target role refinement, which reopens the theorem;
- (iii) a physical target update;
- (iv) derived support, witness, obstruction, or ATS skin rather than an independent role.

There is no theorem that forbids a sixth or twelfth role by vocabulary alone. The theorem is target-relative and remains falsifiable by a same-target missing distinction.

Role/proposal	Disposition	Refinement rule
R9-PRESENTATION	DeclaredRole	A same-target load-bearing burden not governance-equivalent to any declared role is a genuine role-set-refinement counterexample.
R9-COMPARISON	DeclaredRole	A same-target load-bearing burden not governance-equivalent to any declared role is a genuine role-set-refinement counterexample.
R9-REDUCTION	DeclaredRole	A same-target load-bearing burden not governance-equivalent to any declared role is a genuine role-set-refinement counterexample.
R9-RESPONSE	DeclaredRole	A same-target load-bearing burden not governance-equivalent to any declared role is a genuine role-set-refinement counterexample.
R9-CONSISTENCY	DeclaredRole	A same-target load-bearing burden not governance-equivalent to any declared role is a genuine role-set-refinement counterexample.
R9-PROPER-OPERATION	DeclaredRole	A same-target load-bearing burden not governance-equivalent to any declared role is a genuine role-set-refinement counterexample.
R9-BOUNDARY-ACTION	DeclaredRole	A same-target load-bearing burden not governance-equivalent to any declared role is a genuine role-set-refinement counterexample.
R9-GLOBALFORM-TOPOL OGY	DeclaredRole	A same-target load-bearing burden not governance-equivalent to any declared role is a genuine role-set-refinement counterexample.
R9-THETA-REALIZATIO N	DeclaredRole	A same-target load-bearing burden not governance-equivalent to any declared role is a genuine role-set-refinement counterexample.
R9-BRST-REALIZATION	DeclaredRole	A same-target load-bearing burden not governance-equivalent to any declared role is a genuine role-set-refinement counterexample.
R9-SYNCHRONIZATION	DeclaredRole	A same-target load-bearing burden not governance-equivalent to any declared role is a genuine role-set-refinement counterexample.
R9-STAR	Four-way test: exist- ing role / same-target refinement / physical target update / derived support-or-skin	Same-target refinement reopens the theorem; it is not silently absorbed.

4.4 A real automorphism test

The controlled two-chart target has two interchangeable presentation roles and two distinct chart residues. Both residues are eligible for both roles, both bijections are compatible, and a four-element lawful automorphism group acts on the assignments. The two assignments form one orbit. This test demonstrates that the automorphism quotient performs real work rather than serving as a decorative identity group.

Presentation Groupoid, Thin Paths, and Transport Targets

5.1 Three structures that must remain distinct

The gauge architecture requires three distinct groupoid-level objects:

1. the *Čech or presentation groupoid* $\check{\mathcal{G}}_D$, whose objects are local presentations/trivializations and whose arrows are transition maps on overlaps;
2. the *thin path groupoid* $\Pi_1^{\text{thin}}(M)$, whose objects are base points and whose arrows are thin-homotopy classes of paths;
3. the *transport target*, such as the category of G -torsors and equivariant maps, in which parallel transport lives.

Transition functions compare local trivializations on overlaps. Paths connect points of the base. Parallel transport acts between fibers over path endpoints. These are not the same arrow type.[28, 30]

5.2 Boundary-fixed presentation groupoid

Let $\{U_i\}$ be an admitted local cover. A presentation arrow g_{ij} is defined on $U_i \cap U_j$ and obeys

$$g_{ii} = e, \quad g_{ij}^{-1} = g_{ji}, \quad g_{ij}g_{jk} = g_{ik} \quad \text{on } U_i \cap U_j \cap U_k.$$

Restriction to smaller overlaps and partial composition are intrinsic. Boundary conditions restrict which arrows are admitted. The minimal local object is therefore groupoid-like rather than one unqualified global group.[25]

Theorem 5.1 (Boundary-fixed transition groupoid). Admitted local gauge presentations and their boundary-compatible transition maps form a local/restriction groupoid with identity, inverse, partial composition, restriction, and triple-overlap coherence. This theorem fixes the presentation-comparison layer only; it does not by itself construct path transport or a principal bundle with one fixed Lie group.

5.3 Thin path groupoid

A thin path is a smooth path modulo rank-one homotopies that do not sweep area. Composition is concatenation, inversion reverses orientation, and constant paths are identities. This

is the natural domain for smooth parallel-transport functors because a connection's transport is invariant under thin homotopy while retaining curvature-sensitive dependence on genuine loops.[30]

Definition 5.2 (Thin path groupoid). $\Pi_1^{\text{thin}}(M)$ has objects $x \in M$ and arrows $[\gamma] : x \rightarrow y$ given by thin-homotopy classes of smooth paths with sitting instants. Composition is induced by path concatenation.

5.4 Gauge transformations as natural equivalences

For two transport functors PT_A and $\text{PT}_{A'}$, a gauge transformation is represented by a natural equivalence η whose endpoint maps satisfy

$$\eta_y \circ \text{PT}_A([\gamma]) = \text{PT}_{A'}([\gamma]) \circ \eta_x.$$

At the proper-bulk locus, this changes the representative transport functor without changing its physical natural-equivalence class. The operation is skin there. At a boundary, admissible components of η can act on edge data or charges and require a different target record.

Theorem 5.3 (Transition/path/transport separation). The presentation groupoid, thin path groupoid, transport target, and gauge-natural equivalence are distinct typed layers. No theorem may infer the existence of path transport from presentation arrows alone, nor identify a path with a transition map.

5.5 Global-form-indexed transport

A local transition and transport description can be compatible with more than one faithful global gauge group. Every global-sensitive transport target is therefore written $G_\Gamma\text{-Tor}$, with Γ fixed or quantified over the admitted family. Line operators, monopoles, and theta periodicities can distinguish family members. SYM-1 enforces this parameterization but leaves their detailed classification to SYM-9.[2, 32]

Connection as Transport Geometry: Role, Coordinates, Orbit Class, and Variation

6.1 Abstract transport role

Stable endpoint comparison along paths requires a transport assignment compatible with identity paths, inverses, and concatenation. The structural burden is prior to a choice of local one-form. In a Lie-type realization, a connection determines a smooth functor

$$\mathrm{PT}_A : \Pi_1^{\mathrm{thin}}(M) \longrightarrow G\text{-Tor}.$$

The AASC connection paper establishes the necessity of standing-preserving transport comparison; conventional differential geometry supplies one faithful realization.[17, 28, 30]

Theorem 6.1 (Conditional transport realization). If the declared target admits a principal G -bundle or equivalent locally trivial G -torsor geometry and a smooth functorial transport law, then PT_A faithfully realizes the abstract standing-preserving comparison role. The theorem is conditional on the realization data; necessity of comparison alone does not construct the functor for an arbitrary target.

6.2 Four records formerly conflated

6.2.1 Local coordinates

In a local trivialization, a connection is displayed by a Lie-algebra-valued one-form A . The coordinate display is representative-dependent. A change of trivialization gives

$$A \mapsto A^g = gAg^{-1} + g dg^{-1}$$

under the manuscript's convention. The coordinate chart is skin; the connection configuration is not thereby skin.

6.2.2 Vertical gauge variation

A vertical variation moves within a gauge orbit:

$$A \longmapsto A^g, \quad q(A^g) = q(A).$$

At the proper-bulk locus this operation changes no quotient-level physical consequence and is skin.

6.2.3 Physical orbit class

The physical connection configuration is the orbit class

$$[A] \in \mathcal{A}/\mathcal{G}_0.$$

It can carry dynamical field content and cannot be discarded while preserving curvature, holonomy, energy, or amplitudes. The class is typed as a physical tensor/configuration, not as a symmetry action.

6.2.4 Nonvertical or dynamical variation

A variation $A \mapsto A + \delta A$ is not skin when it changes the orbit class. In particular, if

$$\delta F = D_A \delta A \neq 0$$

or a gauge-invariant loop observable changes, the variation moves physical configuration content.

Theorem 6.2 (Connection role split). For the regular proper-bulk target, the following are distinct:

1. the abstract transport-comparison role, a carrier constraint;
2. a local one-form coordinate A , skin as a coordinate display;
3. a vertical transformation $A \mapsto A^g$, skin at the proper quotient locus;
4. the orbit class $[A]$, physical tensor/configuration content;
5. a nonvertical variation of $[A]$, a physical change when invariant consequences differ.

No statement that “the gauge potential is skin” is admitted without specifying which of these objects is meant.

6.3 No canonical-representative authority

A gauge condition or canonical-representative prescription may select one A per orbit locally. If the selection does not change quotient consequences, it is computational skin. If it is used to alter standing or physical status within one orbit, it becomes a non-factorizing selector and is excluded at fixed scope. Global obstructions to such selections are treated in Chapter 12.

Blocked inference

Gauge covariance of A does not imply that arbitrary changes of A are representative-only. Only vertical orbit-preserving changes are skin. A transverse change can alter the physical field configuration.

Curvature, Holonomy, and Wilson Observables

7.1 Curvature as infinitesimal loop residue

For a Lie-type connection,

$$F = dA + A \wedge A, \quad F^g = gFg^{-1}.$$

The local matrix-valued form is gauge covariant. Its invariant contractions and the conjugacy data of small-loop holonomy can carry physical information. The AASC curvature result identifies curvature as the obstruction to path-independent transport and as a boundary-detectable loop residue.[18, 28]

Theorem 7.1 (Curvature role split). Curvature is not a symmetry transformation. In a declared representation it is a physical tensor/configuration datum; gauge-invariant contractions, field equations, and infinitesimal loop response are derived witnesses. Removing curvature from a nonflat target changes transport consequences. Changing only its local basis representative does not.

7.2 Holonomy

For a loop γ based at x ,

$$\text{Hol}_A(\gamma) = \text{PT}_A([\gamma]) \in \text{Aut}(P_x).$$

Under gauge transformation it changes by endpoint conjugation. Its conjugacy class is therefore the natural basepoint-covariant datum. Holonomy is a transport endomorphism or loop residue, not an independent global symmetry operation.

7.3 Wilson loops

Given a representation R ,

$$W_R(\gamma) = \text{Tr}_R \mathcal{P} \exp \oint_{\gamma} A.$$

Taking the trace removes conjugation dependence. The resulting Wilson loop can be gauge invariant.[34] Its physical interpretation depends on the representation, loop class, matter screening, global form, and operator category. For this reason the Wilson-loop record is included as a preview but full line-operator closure is deferred to SYM-9.[2, 10, 32]

Theorem 7.2 (Holonomy/witness non-collapse). Holonomy class and Wilson-loop value are typed as physical tensor/observable data and derived witnesses. They do not receive the role *Physical action* unless a separate transformation record acts on an admitted state or operator sector.

7.4 Flatness and skin

If $F = 0$ on a simply connected region, local transport can be path independent and a connection may be locally pure gauge. This does not imply global triviality: topology, boundary conditions, and holonomy can retain nontrivial data. Conversely, a nonzero local F is not erased by choosing a gauge. The atlas therefore records local curvature, holonomy class, Wilson observable, and topology separately.

Falsifier

A proposed counterexample must preserve the target, representation, loop/operator category, boundary, and global form while showing that an admitted curvature or holonomy distinction changes no quotient-level consequence yet remains independently physical, or changes physical content while the target-role/ATS status analysis returns skin.

Lie-Type Faithful Realization and Product-Sector Compatibility

8.1 No derivation of a fixed Lie group from a bare groupoid

A smooth groupoid with local bisections need not be an action groupoid of one fixed Lie group. It can have varying isotropy, a Lie algebroid, or stack-like structure. The publication version therefore rejects any implication

$$\text{smooth groupoid} \implies \text{unique principal } G\text{-bundle and Yang–Mills data.}$$

The valid result is a conditional realization theorem.

Theorem 8.1 (Lie-type faithful realization). Fix an admitted global form G_Γ . Assume a principal G_Γ -bundle, or an equivalent locally trivial G_Γ -torsor realization, together with smooth functorial parallel transport and compatible local sections. Then local one-forms A_i , transition functions g_{ij} , and curvature forms F_i satisfy the usual compatibility laws and faithfully realize the antecedent transition, transport, and loop-residue roles. No claim is made that those hypotheses are generated from a bare transition groupoid.[19, 28, 30]

Claim status: Conditional faithful-realization theorem.

Falsifier: A target satisfying the stated principal-bundle/torsor and smooth-transport assumptions for which the local data fail the compatibility laws, or a claimed bare-groupoid derivation that avoids adding equivalent realization hypotheses.

8.2 Local gauge product

The Standard Model local Lie algebra/product is conventionally written

$$SU(3)_c \times SU(2)_L \times U(1)_Y.$$

The product notation is structurally licensed only when the sector ledgers coexist in one target: their transition, transport, representation, charge, quotient, and anomaly data must be mutually compatible, while no undeclared mixed residue collapses the factors.[21, 26]

Theorem 8.2 (Product-sector compatibility). Let $\mathfrak{S}_1, \dots, \mathfrak{S}_n$ be proposed gauge sectors. Their product is admitted in one target only if:

1. all sectors share the fixed background, boundary, topology, and operator scope;

2. transition and transport structures compose without conflict;
3. matter responses form compatible product representations;
4. charge and quotient ledgers are jointly defined;
5. all required aggregate gauge anomalies cancel;
6. no observed mixed transport or loop residue requires a different sector decomposition.

8.3 Local product does not fix global form

The local Lie algebra and perturbative point-particle representations can be compatible with quotients by subgroups of the common center. The family

$$G_\Gamma = \frac{SU(3)_c \times SU(2)_L \times U(1)_Y}{\Gamma}$$

changes extended operators and topological sectors. SYM-1 therefore parameterizes every global-sensitive theorem by Γ ; it does not use one unresolved G as a frozen object or infer Γ from the local product.[\[2, 32\]](#)

8.4 Coupling nonclaim

Nothing in the role classification determines g_1, g_2, g_3 , their beta functions, or a unification scale. Couplings are physical parameters and running data downstream of the architecture. Their numerical determination lies outside SYM-1.

Representation Response, Charge Witnesses, and Aggregate Quantum Admissibility

9.1 Individual representation response

A matter representation $\rho_i : G \rightarrow \text{Aut}(V_i)$ specifies how one carrier sector responds to gauge transport and transformations. Classical admissibility requires compatibility with the declared group/algebra, transition structure, covariant derivative, and matter carrier. A basis change

$$\rho_i(g) \mapsto S\rho_i(g)S^{-1}$$

that preserves all response invariants is skin.[22]

Theorem 9.1 (Individual response theorem). An individual representation is classically admissible when it coherently realizes the declared gauge action and carrier response. Its quantum anomaly contribution is a separate diagnostic and need not vanish individually.

9.2 Individual anomaly contribution

For chiral fermions, a representation can contribute

$$\mathcal{A}_i^{abc} = \text{tr}_{\rho_i}(T^a\{T^b, T^c\}),$$

or the relevant Abelian/mixed coefficient. A nonzero $\mathcal{A}_i$ does not by itself make ρ_i unavailable as a matter representation. Quantum consistency is a property of the complete declared chiral ledger.[6]

9.3 Aggregate representation ledger

Let $\mathcal{R} = \bigoplus_i \rho_i$ denote the complete chiral matter ledger with multiplicities and charges. The quantum-gauge target requires cancellation of every relevant local coefficient and satisfaction of global consistency conditions:

$$\sum_i \mathcal{A}_i^{abc} = 0, \quad \sum_i \mathcal{A}_i^{\text{mixed}} = 0,$$

plus any target-relevant global obstruction test. The Standard Model cancellation is an aggregate result.[6, 15, 26, 35]

Theorem 9.2 (Aggregate quantum-ledger admissibility). Individual response admissibility, individual anomaly contribution, and aggregate quantum anomaly neutrality are three distinct records. The aggregate chiral ledger is quantum-admissible only if all target-relevant gauge obstructions cancel. No individual contribution is required to vanish unless the target contains only that representation.

9.4 Charge as response datum, not action

A charge can be an eigenvalue, weight, current integral, or surface functional associated with a generator. It is not itself the transformation. The transformation is

$$U(\alpha) = e^{i\alpha Q}$$

where such an operator realization exists. The atlas therefore types charge as physical tensor/response datum and witness; a separate action record receives the physical-action role.

9.5 Normalization skin

An overall Abelian generator normalization can be traded against the associated coupling convention when all physical products and charge ratios are preserved. That conventional freedom is skin. It does not imply that relative charges, representation weights, or measured responses are conventional.

Blocked inference

“Charge is a convention” is false without qualification. Overall coordinate normalization may be conventional; quotient-stable relative response and measured generator action are physical.

Constraints, Proper Gauge Directions, and Quotient Bridges

10.1 Constraint surface and characteristic distribution

In the canonical description, let C be the constraint surface with pulled-back presymplectic form ω_C . Under regularity, first-class constraints generate characteristic directions. If the proper gauge distribution is maximal and observables separate leaves, then

$$\ker(d\pi) = \ker(\omega_C),$$

The map π is the projection from C to the reduced phase space Q_D^{red} . [7, 12]

Theorem 10.1 (Characteristic quotient identification). Under the declared regularity, differentiability, maximality, and observable-separation assumptions, the proper gauge distribution equals the presymplectic characteristic distribution and the reduced phase-space quotient is well defined. Singular strata, boundary generators, or incomplete observables require a refined target rather than silent inheritance.

10.2 Gauss generator versus global charge

A smeared first-class constraint has the schematic form

$$G[\epsilon] = \int_{\Sigma} \epsilon^a \mathcal{G}_a + \text{boundary term}.$$

For proper parameters vanishing at the boundary, the boundary term vanishes and $G[\epsilon]$ generates a vertical quotient direction. It is not an ordinary global physical charge. For improper parameters, differentiability can require a nonzero surface term; the boundary transformation and charge receive separate records. [12, 29]

10.3 Small-gauge quotient theorem

Let $g \in \mathcal{G}_{0,D}$ be compactly supported or equal to identity on the admitted boundary. Then by definition of the certified proper orbit quotient,

$$q_D^{\text{cl}}(g \cdot x) = q_D^{\text{cl}}(x).$$

Combined with the quotient-fidelity bridge of Theorem 2.2, this action is also standing-neutral for the admitted target.

Theorem 10.2 (Small-gauge quotient theorem). On the regular anomaly-neutral proper-bulk target, a proper compact-support gauge transformation preserving boundary and topology acts trivially on the certified classical quotient. Its operation-level role is skin. This result neither removes the core architecture nor extends to improper, large, topological, or anomalous transformations.

10.4 Factorization after, not before, the bridge

The fixed-domain theorem says that standing-relevant discriminators factor through Q_D^{gov} . The physical conclusion follows only after $Q_D^{\text{gov}} \cong Q_D^{\text{cl}}$ is certified for the target. The logical sequence is therefore

fixed-domain factorization + quotient-fidelity bridge $\implies$ physical proper-gauge factorization.

The anomaly target deliberately blocks this sequence when quantum descent fails.

10.5 Typed quotient registry

The registry below separates governance, classical, cohomological, boundary, and deferred global-form quotients. Its status field controls whether factorization may be transported at a given theorem use site.

Quotient	Target	Domain	Equivalence	Codomain	Status
Q-NONE	D-SYM1-SM-ASSEMBLY	Not applicable	Not applicable	Notapplicable	NotApplicable
Q-GOV-STANDING	D-SYM1-SM-GOVERNANCE	Typed gauge presentations	Equality under all admitted tests after all admitted continuations	Standingclasses	Defined
Q-CL-PROPER-CONFIG	D-SYM1-SM-BULK-PR OPER	Admitted classical gauge configurations	Orbit of proper compact-support transformations preserving boundary data	Classicalreducedconfigurationclasses	DefinedConditionalRegularity
Q-REDUCED-PHASE	D-SYM1-SM-BULK-PR OPER	Constraint surface	Characteristic leaves of the presymplectic form under regularity and maximality	Reducedphasespace	DefinedConditionalRegularity
Q-BOUNDARY-MIXED	D-SYM1-SM-BOUNDARY	Gauge data with declared boundary/asymptotic structure	Proper bulk vertical directions only; boundary action retained when charged	Bulkquotientplusboundarycharge/flux/edgesector	DefinedConditionalBoundaryData
Q-LARGE-DECLARED	D-SYM1-SM-TOP0-QUOTIENT	Fixed topological sector	Includes specified disconnected gauge transformations	Declaredtopologicalquotientclasses	TargetIndexed
Q-PROJECTIVE-RAY	D-SYM1-SM-THETA	Nonzero state vectors	Nonzero complex scalar multiplication	Physicalrays	Defined
Q-BRST-STATE	D-SYM1-SM-BRST-STATE	Extended state complex at ghost number zero	Q_BRST -closed states modulo Q_BRST -exact states	$H^0(Q_BRST, H_{ext})$	DefinedConditionalNilpotency
Q-BRST-LOCAL	D-SYM1-SM-BRST-LOCAL	Local functions/operators on jets, ghosts and antifields	s -closed modulo s -exact at fixed ghost number	$H^*(s, A_{loc})$	DefinedConditionalNilpotency
Q-BRST-RELATIVE	D-SYM1-SM-BRST-RELATIVE	Local n -forms or integrated local functionals	s -closed modulo s -exact and total derivatives d	$H^*\{g, n\}(s d)$	DefinedConditionalNilpotency
Q-QUANTUM-DESCENT	D-SYM1-SM-ANOMALY	Regularized/renormalized quantum gauge representatives	Gauge-related representatives if measure and effective action descend	Quantumgauge-invariantclassesorfailurecertificate	ConditionalMayFail
Q-GLOBALFORM-DEFERRED	D-SYM1-SM-GLOBALFORM-PREVIEW	Extended operator and bundle sectors	Deferred to faithful global form and defect category	Deferred	DeferredSYM9

Proper, Boundary, and Large Gauge Operations

11.1 Why one operation name cannot carry one status

The transformation law can look locally identical while its quotient role changes with boundary, topology, and state sector. The status calculus therefore evaluates operation records by locus rather than by group name.

11.2 Proper bulk operations

At D_{bulk} , the parameter is compactly supported or zero at the admitted boundary. The generator has no physical surface charge, gauge-invariant observables are unchanged, and the transformation lies on a proper gauge orbit. The role is skin at this locus by Theorem 10.2.

11.3 Boundary and asymptotic operations

At D_{∂} , the parameter is admitted to be nonzero at the boundary or asymptotic end. The canonical generator may require a surface term $Q_{\partial}[\epsilon]$ for differentiability. Boundary status is product-valued. The complete status is not one label but

$$\text{BStat}(\epsilon) = (B_{\text{bulk}}, B_{\partial}, Q_{\partial}, \Phi_{\partial}, E_{\partial}).$$

Typical entries are:

- B_{bulk} : vertical or quotient-trivial in the proper bulk;
- B_{∂} : nontrivial action on boundary data or asymptotic states;
- Q_{∂} : zero, nonzero integrable, nonintegrable, or undefined;
- Φ_{∂} : vanishing flux, nonzero flux, or conditional integrability;
- E_{∂} : fixed data, transformed edge data, or target update adding edge modes.

Theorem 11.1 (Boundary product-status theorem). Boundary gauge classification is a product status. Bulk verticality, boundary action, nonzero charge, flux, and edge-data transformation may coexist. A physical-action tag is assigned only to the transformation record with a quotient-level boundary certificate; the surface charge is a generator/value/witness record rather than the action itself.

Claim status: Conditional on declared boundary conditions, differentiable generators, and charge/flux integrability.

Falsifier: A complete boundary setup in which the product fields are insufficient to distinguish two physically different statuses while the target remains fixed.

11.4 Large transformations declared in the quotient

One target may declare specified disconnected transformations as quotient identifications. On that target, they are skin at the operation locus. This is a target convention constrained by global form, topology, and quantum representation; it cannot be inherited by all large transformations.

11.5 Wavefunctional equivariance and projective rays

Suppose

$$\Psi_\theta[A^g] = e^{i\theta\nu(g)}\Psi_\theta[A].$$

If this is only multiplication of one state vector by a common nonzero complex number, it acts trivially on the projective ray. It may encode an equivariance condition or choice of quantization character without being a physical action on that ray.

Theorem 11.2 (Phase insufficiency). A wavefunctional phase under a large transformation is not sufficient for the role *Physical action*. A separate certificate must show a change of physical ray, superselection sector, operator algebra, interference observable, partition function, or other quotient-level datum.

11.6 Certified theta-sector action

A large transformation can act physically when it changes a sector label, acts nontrivially on an admitted operator, or has an observable relation between sectors. The theta datum itself is an anchor of the selected quantization target. Changing it is not a same-scope operation unless the target includes theory-space variation.

Theorem 11.3 (Large-gauge/theta split). The atlas separates: large transformations included in the quotient; wavefunctional equivariance; theta-sector target data; conditional sector/operator actions; fixed topology; and topology updates. No status is inherited from the phrase “large gauge transformation” alone.[2, 33]

11.7 Residual transformations

Residual transformations left after a gauge condition may be proper redundancy, boundary action, Gribov-related multiplicity, or a physical global stabilizer. Their status follows from

the unfixed target and quotient, not from the fact that they preserve the gauge condition. Electroweak residual $U(1)_{\text{em}}$ is deferred to SYM-8 because it requires vacuum fixation and spectrum reconstruction.

Gauge Fixing, BRST Realization, and Three Cohomological Targets

12.1 Gauge fixing has no independent physical authority

A gauge condition $\chi[A] = 0$ selects or weights representatives. In perturbation theory, the Faddeev–Popov procedure introduces a determinant and ghost fields.[8] These structures can be indispensable to a chosen quantization while remaining downstream of the physical quotient they represent.

Theorem 12.1 (Gauge-fixing no-authority theorem). Within a fixed gauge target, a gauge condition cannot become a second physical quotient or canonical standing authority. If it changes no quotient consequence it is skin; if it changes which physical states or observables are admitted, it changes target or exposes a failure of the proposed gauge fixing.

12.2 Gribov and Singer obstructions

Non-Abelian gauge theories generally do not admit a single global smooth gauge-fixing section under broad conditions. Gribov copies and Singer’s topological obstruction demonstrate that a local representative prescription cannot be promoted to a globally canonical physical structure.[11, 31] The obstruction is a witness against global gauge fixing, not evidence that the physical quotient is illusory.

12.3 BRST realization architecture

The BRST target contains gauge-fixing data, ghosts and auxiliaries. It also contains a nilpotent differential or charge, ghost-number grading, and a typed cohomological projection.[4, 12] It is a target-relative realization architecture:

$$\mathfrak{A}_{\text{gauge}}^{\text{BRST}} = (\chi, c, \bar{c}, b, s, Q_{\text{BRST}}, H_{\text{state}}, H_{\text{local}}, H_{\text{rel}}).$$

It is a faithful realization of the physical-reduction and quantum-consistency obligations in $\mathfrak{R}_D^{\text{gauge}}$, not a universal primitive of that role core.

12.4 State cohomology

For a nilpotent BRST charge Q_{BRST} acting on an extended state space,

$$\mathcal{H}_{\text{phys}} \simeq H^0(Q_{\text{BRST}}, \mathcal{H}_{\text{ext}}) = \frac{\ker Q_{\text{BRST}} \cap \mathcal{H}_{\text{gh}=0}}{\text{im } Q_{\text{BRST}} \cap \mathcal{H}_{\text{gh}=0}}.$$

The exact denominator arises from states of ghost number -1 . Inner-product and completion assumptions are target-specific. Global-obstruction assumptions are target-specific as well.[12, 13]

12.5 Local operator cohomology

For local functions or operators on the chosen jet/field/antifield algebra,

$$H^g(s, \mathcal{A}_{\text{loc}}) = \frac{\ker(s : \mathcal{A}_{\text{loc}}^g \rightarrow \mathcal{A}_{\text{loc}}^{g+1})}{\text{im}(s : \mathcal{A}_{\text{loc}}^{g-1} \rightarrow \mathcal{A}_{\text{loc}}^g)}.$$

This classifies local gauge-invariant content under the specified on-shell/off-shell and antifield conventions. It is not the physical Hilbert space.[3]

12.6 Relative local cohomology

Integrated local functionals, consistent deformations, currents, and local anomaly candidates require cohomology modulo total derivatives:

$$H^{g,n}(s \mid d).$$

The consistency condition has the descent form $sa + db = 0$, with equivalence $a \sim a + sm + dn$. This target is neither state cohomology nor ordinary local $H^g(s)$.

Theorem 12.2 (BRST three-complex theorem). BRST state cohomology, local operator cohomology, and relative local cohomology are distinct typed objects with distinct domains, equivalence relations, and physical uses. No one expression $H^0(s)$ may substitute for all three.

12.7 Ghost fields and ghost number

Ghost fields are representative degrees of freedom in the gauge-fixed complex, not ordinary asymptotic physical particles in the declared target. Their determinant and cohomological role are derived quantization witnesses. Ghost number is a grading of the complex, not an ordinary Standard Model particle charge.

12.8 Target-relative minimality

Removing BRST from the formulation-independent role core does not remove any role obligation. Removing the nilpotent differential or complex from D_{BRST} destroys that realization target. Thus:

Theorem 12.3 (BRST target-relative minimality). BRST components are minimal only relative to the declared BRST realization target. No formulation-specific BRST component is promoted into the formulation-independent gauge-role core.

Nilpotency condition

The cohomological bridges require anomaly-free nilpotency and compatible boundary conditions. If $s^2 \neq 0$ or $Q_{\text{BRST}}^2 \neq 0$ because quantum gauge descent fails, the bridge is blocked rather than repaired by notation.

Gauge Anomaly as Quantum-Descent Failure

13.1 Classical invariance does not guarantee quantum descent

A classical symmetry of the action need not survive quantization. Adler and Bell–Jackiw established the anomalous axial-current divergence through perturbative triangle-diagram analyses.[1, 5] Fujikawa identified the corresponding non-invariance of the fermionic path-integral measure under a chiral transformation.[9] For chiral gauge theories, the effective-action, consistency, descent, and BRST-cohomological formulations distinguish a genuine gauge anomaly from an anomalous global current.[3, 6]

13.2 Local perturbative anomaly

For a chiral representation ledger, the consistent local anomaly is governed by the relevant invariant polynomial. In BRST language a local anomaly candidate lies in ghost number one relative cohomology, schematically

$$[a] \in H^{1,n}(s \mid d).$$

If the class is nontrivial and its aggregate coefficient does not cancel, the effective action does not descend to gauge-equivalence classes.

Theorem 13.1 (Gauge-anomaly descent theorem). A noncanceled gauge anomaly is a failure of quantum gauge descent, not an allowed explicitly broken gauge symmetry and not a spontaneous vacuum effect. The physical quantum quotient is therefore undefined or inconsistent for that target. The SYM-0 status is *Obstruction*; it does not assume factorization through the quotient it is testing.

13.3 Individual contribution versus aggregate obstruction

Each chiral multiplet can contribute a nonzero coefficient. The aggregate ledger sums all contributions with multiplicity and charge. Quantum admissibility requires cancellation of all target-relevant local gauge, mixed gauge, and gauge-gravitational coefficients. The representation itself and its contribution remain valid diagnostic records even when the aggregate cancels.

13.4 Global gauge anomaly

The Witten $SU(2)$ anomaly is a nonperturbative global obstruction not visible in the local cubic polynomial.[35] It depends on global group and representation data. SYM-1 includes the Standard Model parity check as an obstruction/witness record but defers a complete faithful-global-form audit to SYM-9.

13.5 Counterterms and inflow

A local counterterm can shift an anomaly representative between consistent forms but cannot remove a nontrivial obstruction unless the relevant cohomology class is trivial. An inflow sector can cancel a boundary anomaly only by enlarging the target to include the bulk or additional degrees of freedom. Within a fixed target, later addition of such a sector is a target update, not retrospective repair.

13.6 Standard Model anomaly-neutrality ledger

The Standard Model matter content satisfies the perturbative gauge-anomaly cancellation conditions generation by generation in the usual local target. The corpus records the relation between product structure, representations, charges, and anomaly neutrality without claiming to derive numerical couplings.[6, 15, 26]

Theorem 13.2 (Aggregate anomaly-neutrality condition). The formulation-independent quantum gauge target is admitted only when the complete declared representation and charge ledger is neutral under every required gauge obstruction. Individual coefficients are witnesses; their aggregate cancellation is the carrier constraint; a nonzero aggregate is an obstruction.

No anomaly circularity

A valid objection may show that a source coefficient, representation multiplicity, global-group assumption, or descent classification is wrong. It may not assume a physical quantum quotient in the anomalous target and then criticize the manuscript for failing to factor through it; the target explicitly records descent failure.

Integrated Candidate Generation, Presentation Residue, and Eligibility

14.1 Target-derived protocol

The candidate generator traverses the primitive interfaces and relations of the integrated target. Its families are fixed before atlas traversal.

Family	Name	N	Target traversal fields
V9-PF-PRESENTATION	Local presentation and overlap coherence	7	local_presentations; transition_maps; restriction_maps
V9-PF-TRANSPORT	Comparison, connection, and transport	11	paths; connections; parallel_transport; holonomy
V9-PF-REDUCTION	Constraint and physical quotient	5	constraint_surface; proper_gauge_quotient; quotient_bridge
V9-PF-RESPONSE	Matter representation and charge response	2	representation_ledger; charge_witnesses
V9-PF-CONSISTENCY	Aggregate quantum gauge consistency	8	anomaly_contributions; aggregate_anomaly_ledger; global_anomaly_check
V9-PF-PROPER	Proper compact-support operation class	1	proper_local_operations; vertical_connection_variations
V9-PF-BOUNDARY	Boundary/asymptotic operation package	7	boundary_actions; surface_charges; integrability; flux
V9-PF-GLOBALFORM	Global-form and topology package	2	global_form; bundle_sector; topology; Wilson_observables
V9-PF-THETA	Large-transformation and theta representation	8	large_transformations; equivariance; theta_character; sector_action
V9-PF-BRST	Gauge-fixing and BRST/BV realization	11	gauge_fixing; ghost_complex; BRST_differential; state_local_relative_cohomology
V9-PF-SYNC	Cross-realization synchronization	1	geometric_module; Hamiltonian_module; BRST_module; global_form; representation; quotient; anomaly
V9-PF-DIAGNOSTIC	Derived witnesses and obstruction records	9	Ward_identities; Wilson_witnesses; anomaly_diagnostics; Gribov_diagnostics
V9-PF-TARGET-EDGE	Explicit target extensions and downstream handoffs	3	new_fields; new_global_form; new_topology; vacuum_handoff; line_operator_handoff

14.2 Candidate universe

The generator emits atlas-backed target views, composite packages assembled from declared target relations, atlas-external diagnostics, and presentation-equivalent stress records. Adding a declared target object changes the generated universe without editing the atlas.

Target	All	Atlas	New	Dup.	Function
D-SYM1-SM-ASSEMBLY	73	57	16	5	Primary integrated SYM-1 closure target: coherent presentation, comparison/transport, physical reduction, matter response, quantum consistency, proper and boundary operations, topology/global form, theta realization, BRST realization, and cross-realization synchronization.
D-SYM1-TWO-CHART-PRESENTATION	2	0	2	0	Controlled nontrivial certificate-lawful automorphism test: two chart roles and two presentation residues form two assignments in one S2 orbit.

Test	Result	Meaning
V9-PROT-ATLAS-EXTERNAL	PASS	Target-declared stabilizer field emits an atlas-external candidate.
V9-PROT-PRESENTATION-DUPPLICATES	PASS	Five gauge-theoretic presentation duplicates are generated independently of atlas rows.
V9-PROT-SAME-TARGET-GAP	PASS	A declared primitive interface absent from the generator is classified as a theorem-breaking candidate gap.

Theorem 14.1 (Target-signature candidate coverage). Every candidate expressible through the frozen integrated SYM-1 target signature is emitted by one protocol family. A gauge-relevant same-target object present in the target but absent from the generator is a candidate-completeness counterexample and reopens the theorem.

Claim status: Finite representation theorem relative to the declared integrated target signature; not a claim about arbitrary ultraviolet completions or SYM-9 defect completion.

14.3 Presentation quotient

The quotient is tested on multiple gauge-theoretic presentation pairs rather than one synthetic duplicate.

Presentation-quotient object	Result
Candidate-locus records	75
Presentation classes	70
Non-singleton gauge-theoretic classes	5
Equivalence-relation laws	PASS
Eligibility and compatibility invariance inside each class	PASS

Presentation class	Members
PC9-SYM-01-BRST-LOCAL-COHOMOLOGY-v2	SYM-01-BRST-LOCAL-COHOMOLOGY-v2@@D-SYM1-SM-ASSEMBLY; V9-DUP-BRST-EXACT-REP
PC9-SYM-01-REPRESENTATION-RESPONSE-v1	SYM-01-REPRESENTATION-RESPONSE-v1; V9-DUP-REPRESENTATION-WEAK-BASIS
PC9-SYM-01-THETA-SECTOR-DATUM-v2	SYM-01-THETA-SECTOR-DATUM-v2@@D-SYM1-SM-ASSEMBLY; V9-DUP-THETA-PHASE-CONVENTION
PC9-SYM-01-TRANSITION-GROUPOID-v1	SYM-01-TRANSITION-GROUPOID-v1; V9-DUP-TRANSITION-RELABELLED
PC9-SYM-01-TRANSPORT-FUNCTOR-v2	SYM-01-TRANSPORT-FUNCTOR-v2; V9-DUP-TRANSPORT-NATURAL

Theorem 14.2 (Presentation-quotient soundness). Members of one presentation class have identical role capability, compatibility profile, target, locus, quotient, boundary, topology, representation, anomaly, continuation, and failure invariants. Every retained pair has at least one certified surviving distinction. Consequently the quotient collapses only presentation-level duplication within the frozen certificate.

14.4 Full eligibility relation

Every candidate-role pair is evaluated from the candidate's certified capabilities and the occupant-independent role criterion. The resulting positive graph is nontrivial.

Target	Role	N	Eligible alternatives
D-SYM1-SM-ASSEMBLY	R9-BOUNDARY-ACTION	2	RES9-V9-PKG-BOUNDARY; RES9-V9-PKG-BOUNDARY-NONINTEGRABLE
D-SYM1-SM-ASSEMBLY	R9-BRST-REALIZATION	2	RES9-SYM-01-BRST-REALIZATION-ARCH-v2; RES9-V9-PKG-BRST
D-SYM1-SM-ASSEMBLY	R9-COMPARISON	2	RES9-SYM-01-CONNECTION-ROLE-v1; RES9-SYM-01-TRANSPORT-FUNCTOR-v2
D-SYM1-SM-ASSEMBLY	R9-CONSISTENCY	2	RES9-SYM-01-ANOMALY-NEUTRALITY-v1; RES9-V9-PKG-CONSISTENCY
D-SYM1-SM-ASSEMBLY	R9-GLOBALFORM-TOPOLOGY	2	RES9-SYM-01-GLOBAL-FORM-PLACEHOLDER-v1; RES9-V9-PKG-GLOBALFORM-TOPOLOGY
D-SYM1-SM-ASSEMBLY	R9-PRESENTATION	2	RES9-SYM-01-TRANSITION-GROUPOID-v1; RES9-V9-PKG-PRESENTATION
D-SYM1-SM-ASSEMBLY	R9-PROPER-OPERATION	2	RES9-SYM-01-CONNECTION-VERTICAL-VARIATION-v2; RES9-SYM-01-GAUGE-LOCAL-PROPER-v1
D-SYM1-SM-ASSEMBLY	R9-REDUCTION	3	RES9-SYM-01-CLASSICAL-GAUGE-QUOTIENT-v2; RES9-SYM-01-GOVERNANCE-QUOTIENT-v2; RES9-V9-PKG-REDUCTION
D-SYM1-SM-ASSEMBLY	R9-RESPONSE	2	RES9-SYM-01-CHARGE-WITNESS-v1; RES9-SYM-01-REPRESENTATION-RESPONSE-v1
D-SYM1-SM-ASSEMBLY	R9-SYNCHRONIZATION	2	RES9-SYM-01-SM-ASSEMBLY-CERT-v1; RES9-V9-PKG-SYNCHRONIZATION
D-SYM1-SM-ASSEMBLY	R9-THETA-REALIZATION	3	RES9-SYM-01-GAUGE-LARGE-SECTOR-ACTION-RULE-v2; RES9-SYM-01-THETA-SECTOR-DATUM-v2; RES9-V9-PKG-THETA
D-SYM1-TWO-CHART-PRESENTATION	R9-CHART-A	2	RES9-CHART-A; RES9-CHART-B
D-SYM1-TWO-CHART-PRESENTATION	R9-CHART-B	2	RES9-CHART-A; RES9-CHART-B

Theorem 14.3 (Eligibility completeness). The active closure targets contain no unresolved residue-role pair. Positive edges arise from target-internal capability typing; negative edges identify the missing criterion. The intended complete assignment is computed only after the matrix is frozen.

Joint Compatibility, Live Residue, and Assignment Orbits

15.1 All pointwise eligible preassignments

The integrated incidence graph produces 4,608 injective pointwise eligible preassignments. Every one is evaluated. The compatibility coordinates are calculated from occupant profiles rather than initialized to acceptance.

Integrated compatibility object	Count
Pointwise eligible injective preassignments	4608
Jointly compatible preassignments	1
Pointwise eligible residue	24
Live residue	11
Complete assignments	1

Failed compatibility component	Rejected
quotient_crosscheck	4224
theta_physicality	3840
globalform_theta_crosscheck	3840
aggregate_anomaly_consistency	3456
response_anomaly_crosscheck	3456
boundary_topology_crosscheck	3456
physical_reduction	3072
presentation_coherence	2304
transport_completeness	2304
matter_response	2304
proper_operation_scope	2304
globalform_topology	2304
brst_realization	2304
cross_realization_synchronization	2304
boundary_integrability	2304

Theorem 15.1 (Independent joint-compatibility calculation). For each pointwise eligible preassignment, the target, presentation, transport, reduction, representation, anomaly, proper-operation, boundary, global-form/topology, theta, BRST, synchronization, and cross-interface predicates are evaluated independently. Exactly one integrated preassignment satisfies every coordinate.

15.2 Eligible and live residue

The live residue is not selected from the intended assignment. It is defined by

$$X_D^{\text{live}} = \bigcup_{\tau \in \Sigma_D^{\text{pre}}, J_D(\tau)=1} \text{im}(\tau).$$

Pointwise eligible alternatives absent from every compatible assignment receive explicit failure diagnoses.

Class	Residue identifiers
D-SYM1-SM-ASSEMBLY: eligible	RES9-SYM-01-ANOMALY-NEUTRALITY-v1; RES9-SYM-01-BRST-REALIZATION-ARCH-v2; RES9-SYM-01-CHARGE-WITNESS-v1; RES9-SYM-01-CLASSICAL-GAUGE-QUOTIENT-v2; RES9-SYM-01-CONNECTION-ROLE-v1; RES9-SYM-01-CONNECTION-VERTICAL-VARIATION-v2; RES9-SYM-01-GAUGE-LARGE-SECTOR-ACTION-RULE-v2; RES9-SYM-01-GAUGE-LOCAL-PROPER-v1; RES9-SYM-01-GLOBAL-FORM-PLACEHOLDER-v1; RES9-SYM-01-GOVERNANCE-QUOTIENT-v2; RES9-SYM-01-REPRESENTATION-RESPONSE-v1; RES9-SYM-01-SM-ASSEMBLY-CERT-v1; RES9-SYM-01-THETA-SECTOR-DATUM-v2; RES9-SYM-01-TRANSITION-GROUPOID-v1; RES9-SYM-01-TRANSPORT-FUNCTOR-v2; RES9-V9-PKG-BOUNDARY; RES9-V9-PKG-BOUNDARY-NONINTEGRABLE; RES9-V9-PKG-BRST; RES9-V9-PKG-CONSISTENCY; RES9-V9-PKG-GLOBALFORM-TOPOLOGY; RES9-V9-PKG-PRESENTATION; RES9-V9-PKG-REDUCTION; RES9-V9-PKG-SYNCHRONIZATION; RES9-V9-PKG-THETA
D-SYM1-SM-ASSEMBLY: live	RES9-SYM-01-GAUGE-LOCAL-PROPER-v1; RES9-SYM-01-REPRESENTATION-RESPONSE-v1; RES9-SYM-01-TRANSPORT-FUNCTOR-v2; RES9-V9-PKG-BOUNDARY; RES9-V9-PKG-BRST; RES9-V9-PKG-CONSISTENCY; RES9-V9-PKG-GLOBALFORM-TOPOLOGY; RES9-V9-PKG-PRESENTATION; RES9-V9-PKG-REDUCTION; RES9-V9-PKG-SYNCHRONIZATION; RES9-V9-PKG-THETA
D-SYM1-TWO-CHART-PRESENTATION: eligible	RES9-CHART-A; RES9-CHART-B
D-SYM1-TWO-CHART-PRESENTATION: live	RES9-CHART-A; RES9-CHART-B

Target	Non-live residue	Failed components
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-ANOMALY-NEUTRALITY-v1	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-BRST-REALIZATION-ARCH-v2	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-CHARGE-WITNESS-v1	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness

Target	Non-live residue	Failed components
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-CLASSICAL-GAUGE-QUOTIENT-v2	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-CONNECTION-ROLE-v1	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-CONNECTION-VERITICAL-VARIATION-v2	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-GAUGE-LARGE-SECTOR-ACTION-RULE-v2	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-GLOBAL-FORM-PLACEHOLDER-v1	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-GOVERNANCE-QUOTIENT-v2	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-SM-ASSEMBLY-CERT-v1	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-THETA-SECTOR-DATUM-v2	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness

Target	Non-live residue	Failed components
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-TRANSITION-GROUP UPOID-v1	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-V9-PKG-BOUNDARY-NONINTEGRABLE	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness

15.3 Complete assignments

A complete assignment is a bijection from the target role set onto the computed live residue satisfying eligibility and joint compatibility. The integrated assignment is:

Target role	Live occupant
R9-PRESENTATION	RES9-V9-PKG-PRESENTATION
R9-COMPARISON	RES9-SYM-01-TRANSPORT-FUNCTOR-v2
R9-REDUCTION	RES9-V9-PKG-REDUCTION
R9-RESPONSE	RES9-SYM-01-REPRESENTATION-RESPONSE-v1
R9-CONSISTENCY	RES9-V9-PKG-CONSISTENCY
R9-PROPER-OPERATION	RES9-SYM-01-GAUGE-LOCAL-PROPER-v1
R9-BOUNDARY-ACTION	RES9-V9-PKG-BOUNDARY
R9-GLOBALFORM-TOPOLOGY	RES9-V9-PKG-GLOBALFORM-TOPOLOGY
R9-THETA-REALIZATION	RES9-V9-PKG-THETA
R9-BRST-REALIZATION	RES9-V9-PKG-BRST
R9-SYNCHRONIZATION	RES9-V9-PKG-SYNCHRONIZATION

15.4 Certificate-lawful automorphisms and orbits

The integrated invariant tuple rigidly distinguishes its roles and live occupants, so its lawful automorphism group is the identity. The two-chart target has a four-element lawful group generated by independent role and residue swaps. Its two complete assignments lie in one orbit.

Target	Assign.	$ \Gamma $	Orbits	Closed	Orbit characterization
D-SYM1-SM-ASSEMBLY	1	1	1	yes	Rigid integrated assignment.
D-SYM1-TWO-CHART-PRESENTATION	2	4	1	yes	Two chart assignments identified by the four-element lawful incidence group.

Theorem 15.2 (Integrated Gauge Architecture Occupancy Closure). The integrated Standard Model gauge target has one complete compatible assignment and one lawful assignment orbit. The controlled two-chart target has two assignments and one lawful orbit under a

nontrivial automorphism group. Thus the role-occupancy quotient is singleton in both the rigid and genuinely symmetric cases.

Proof. The role sets are fixed independently of residue. The target-derived candidate universe is quotiented only by certified presentation equivalence. Every residue-role pair is classified. Every injective eligible map is tested by the full compatibility predicate. Live residue is computed from compatible images. Every bijection from roles onto live residue is enumerated. Finally, all automorphisms preserving the complete finite invariant tuple are applied to the assignment set. The generated registries give one orbit for each active closure target. $\square$

Claim status: Exhaustive finite model-checking theorem relative to the frozen integrated certificate, combined with the imported Role-Occupancy Closure theorem.

15.5 Cardinality is only the shadow

The equality $|R_D| = |X_D^{\text{live}}|$ follows from the complete bijection. It is not used to establish closure. Incidence, compatibility, and lawful orbit uniqueness remain essential.

Certificate Completeness and Conditional Integrated Closure

16.1 Seven completeness clauses

The sufficiency direction of Role-Occupancy Closure is invoked only after:

- (i) candidate completeness relative to the frozen target signature;
- (ii) presentation-quotient completeness relative to the certified invariant fingerprint;
- (iii) occupant-independent role-set completeness;
- (iv) total eligibility classification;
- (v) evaluation of every eligible preassignment under all declared compatibility components;
- (vi) live-residue completeness;
- (vii) failure completeness.

Target	Cand.	Quot.	Roles	Elig.	Compat.	Live	Fail	Endpoint
D-SYM1-SM-ASSEMBLY	P	P	P	P	P	P	P	Closed (relative)
D-SYM1-TWO-CHART-PRESENTATION	P	P	P	P	P	P	P	Closed (relative)

Theorem 16.1 (Role-realization completeness). The integrated target and the two-chart automorphism target satisfy all seven completeness clauses relative to their frozen certificates. Each has at least one complete assignment and exactly one lawful assignment orbit.

Claim status: Certificate-relative finite theorem. A same-target missing role, primitive interface, candidate, eligibility edge, compatibility condition, or invariant remains an explicit falsifier.

16.2 Conditional integrated gauge closure

Theorem 16.2 (Conditional fixed-target gauge closure). Relative to the frozen integrated SYM-1 certificate, realization of the declared Standard Model gauge target is licensed. No generated same-target candidate remains untyped: every candidate is presentation-equivalent to a classified residue and is either a live occupant, a compatibility-impossible

eligible alternative, an ineligible peripheral record, a derived ATS/SYM-0 witness or obstruction, or an explicit target edge.

Proof. Apply Theorems 15.2 and 16.1 and the imported Role-Occupancy Closure biconditional Theorem 1.2. The certificate-relative boundary is preserved by the same-target gap and role-refinement rules. $\square$

16.3 Three-axis closure

Every active atlas row carries an independent occupancy disposition, ATS type, and SYM-0 status vector.

Record	Target-role occupancy	ATS	SYM-0 status
SYM-01-GOVERNANCE-QUOTIENT-v2	None	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=NotApplicable;anomaly=NotApplicable
SYM-01-QUOTIENT-FIDELITY-BRIDGE-v2	R9-REDUCTION(ConstituentOfPackage)	DerivedWitness	q=NotApplicable;disposition=QuotientBridge;quantum=NotApplicable;anomaly=None
SYM-01-CHARGE-NORMALIZATION-v1	None	Skin	q=Trivial;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-CHARGE-WITNESS-v1	None	PhysicalTensor;DerivedWitness	q=Nontrivial;disposition=PhysicalTensorDatum;quantum=Exact;anomaly=None
SYM-01-CONNECTION-ROLE-v1	None	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-CURVATURE-ROLE-v1	None	CarrierConstraint;DerivedWitness	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-GAUGE-ARCH-v1	R9-SYNCHRONIZATION(ConstituentOfPackage)	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-HOLONOMY-CLASS-v1	None	PhysicalTensor;DerivedWitness	q=Nontrivial;disposition=PhysicalTensorDatum;quantum=Exact;anomaly=None
SYM-01-LIE-SPECIALIZATION-v1	R9-SYNCHRONIZATION(ConstituentOfPackage)	DerivedWitness	q=NotApplicable;disposition=StructuralDatum;quantum=Conditional;anomaly=None
SYM-01-PRODUCT-COMPATIBILITY-v1	R9-SYNCHRONIZATION(ConstituentOfPackage)	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-REPRESENTATION-RESPONSE-v1	R9-RESPONSE(DirectOccupant)	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=Conditional;anomaly=None
SYM-01-SM-ASSEMBLY-CERT-v1	R9-SYNCHRONIZATION(ConstituentOfPackage)	DerivedWitness	q=NotApplicable;disposition=DiagnosticNotAction;quantum=Exact;anomaly=None
SYM-01-SM-LOCAL-PRODUCT-v1	R9-PRESENTATION(ConstituentOfPackage)	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-TRANSITION-GROUPOID-v1	R9-PRESENTATION(ConstituentOfPackage)	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-GAUGE-NATURAL-EQUIVALENCE-v2	None	Skin	q=Trivial;disposition=TrivialAction;quantum=NotApplicable;anomaly=None
SYM-01-THIN-PATH-GROUPOID-v2	None	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=NotApplicable;anomaly=None
SYM-01-TRANSPORT-FUNCTOR-v2	R9-COMPARISON(DirectOccupant)	CarrierConstraint	q=Mixed;disposition=StructuralDatum;quantum=NotApplicable;anomaly=None
SYM-01-CLASSICAL-GAUGE-QUOTIENT-v2	R9-REDUCTION(ConstituentOfPackage)	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=NotApplicable;anomaly=None
SYM-01-CONNECTION-DYNAMICAL-VARIATION-v2	None	PhysicalTensor	q=Nontrivial;disposition=PhysicalTensorDatum;quantum=Conditional;anomaly=None
SYM-01-CONNECTION-LOCAL-COORDINATES-v2	None	Skin	q=Trivial;disposition=StructuralDatum;quantum=NotApplicable;anomaly=None
SYM-01-CONNECTION-ORBIT-CLASS-v2	None	PhysicalTensor	q=Nontrivial;disposition=PhysicalTensorDatum;quantum=Conditional;anomaly=None
SYM-01-CONNECTION-VERTICAL-VARIATION-v2	None	Skin	q=Trivial;disposition=TrivialAction;quantum=Conditional;anomaly=None
SYM-01-CONSTRAINT-SURFACE-v1	R9-REDUCTION(ConstituentOfPackage)	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-CURVATURE-FORM-v1	None	PhysicalTensor;DerivedWitness	q=Mixed;disposition=PhysicalTensorDatum;quantum=Exact;anomaly=None
SYM-01-GAUGE-LOCAL-PROPER-v1	R9-PROPER-OPERATION(DirectOccupant)	Skin	q=Trivial;disposition=TrivialAction;quantum=Exact;anomaly=None
SYM-01-GAUSS-GENERATOR-PROPER-v1	R9-REDUCTION(ConstituentOfPackage)	DerivedWitness	q=Trivial;disposition=GeneratorNotAction;quantum=Exact;anomaly=None
SYM-01-REPRESENTATION-BASIS-v1	None	Skin	q=Trivial;disposition=TrivialAction;quantum=Exact;anomaly=None
SYM-01-GAUGE-BOUNDARY-ACTION-v1	R9-BOUNDARY-ACTION(ConstituentOfPackage)	PhysicalAction	q=Mixed;disposition=NontrivialAction;quantum=Conditional;anomaly=None
SYM-01-GAUGE-BOUNDARY-RULE-v1	R9-BOUNDARY-ACTION(ConstituentOfPackage)	DerivedWitness	q=Mixed;disposition=DiagnosticNotAction;quantum=Conditional;anomaly=None
SYM-01-SURFACE-CHARGE-v1	R9-BOUNDARY-ACTION(ConstituentOfPackage)	PhysicalTensor;DerivedWitness	q=Nontrivial;disposition=PhysicalTensorDatum;quantum=Exact;anomaly=None

Record	Target-role occupancy	ATS	SYM-0 status
SYM-01-GAUGE-LARGE-EQUIVARIANCE-v2	R9-THETA-REALIZATION (ConstituentOfPackage)	DerivedWitness	q=Trivial;disposition=DiagnosticNotAction;quantum=Conditional;anomaly=None
SYM-01-GAUGE-LARGE-QUOTIENT-v1	None	Skin	q=Trivial;disposition=TrivialAction;quantum=Exact;anomaly=None
SYM-01-GAUGE-LARGE-SECTOR-ACTION-RULE-v2	R9-THETA-REALIZATION (ConstituentOfPackage)	PhysicalAction	q=Mixed;disposition=ConditionalActionRule;quantum=Conditional;anomaly=None
SYM-01-THETA-SECTOR-DATUM-v2	R9-THETA-REALIZATION (ConstituentOfPackage)	Anchor	q=NotApplicable;disposition=StructuralDatum;quantum=Conditional;anomaly=None
SYM-01-TPOLOGY-ANCHOR-v1	R9-GLOBALFORM-TPOLOGY (ConstituentOfPackage);R9-THETA-REALIZATION (ConstituentOfPackage)	Anchor	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-TPOLOGY-UPDATE-v1	None	TargetUpdate	q=NotApplicable;disposition=TargetUpdateDisposition;quantum=Exact;anomaly=None
SYM-01-BRST-DIFFERENTIAL-v1	R9-BRST-REALIZATION (ConstituentOfPackage)	DerivedWitness	q=Mixed;disposition=CohomologicalProjection;quantum=PerturbativelyExact;anomaly=BRST
SYM-01-BRST-REALIZATION-ARCH-v2	R9-BRST-REALIZATION (ConstituentOfPackage)	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=PerturbativelyExact;anomaly=BRST
SYM-01-FP-GHOSTS-v1	R9-BRST-REALIZATION (ConstituentOfPackage)	Skin;DerivedWitness	q=Trivial;disposition=StructuralDatum;quantum=PerturbativelyExact;anomaly=BRST
SYM-01-GAUGE-FIXING-v1	R9-BRST-REALIZATION (ConstituentOfPackage)	Skin	q=Trivial;disposition=StructuralDatum;quantum=PerturbativelyExact;anomaly=None
SYM-01-GHOST-NUMBER-v1	None	DerivedWitness	q=NotApplicable;disposition=DiagnosticNotAction;quantum=Conditional;anomaly=BRST
SYM-01-GRIBOV-OBSTRUCTION-v1	None	Obstruction;DerivedWitness	q=Nontrivial;disposition=DiagnosticNotAction;quantum=Conditional;anomaly=None
SYM-01-BRST-STATE-COHOMOLOGY-v2	R9-BRST-REALIZATION (ConstituentOfPackage)	DerivedWitness	q=Nontrivial;disposition=CohomologicalProjection;quantum=PerturbativelyExact;anomaly=BRST
SYM-01-BRST-LOCAL-COHOMOLOGY-v2	R9-BRST-REALIZATION (ConstituentOfPackage)	DerivedWitness	q=Nontrivial;disposition=CohomologicalProjection;quantum=PerturbativelyExact;anomaly=BRST
SYM-01-BRST-RELATIVE-COHOMOLOGY-v2	R9-BRST-REALIZATION (ConstituentOfPackage)	DerivedWitness	q=Nontrivial;disposition=CohomologicalProjection;quantum=PerturbativelyExact;anomaly=BRST
SYM-01-AGGREGATE-REP-LEDGER-v2	R9-CONSISTENCY (ConstituentOfPackage)	CarrierConstraint;DerivedWitness	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-ANOMALY-NEUTRALITY-v1	R9-CONSISTENCY (ConstituentOfPackage)	CarrierConstraint;DerivedWitness	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-INFLOW-UPDATE-v1	None	TargetUpdate	q=NotApplicable;disposition=TargetUpdateDisposition;quantum=Conditional;anomaly=Unresolved
SYM-01-LOCAL-GAUGE-ANOMALY-v1	None	Obstruction	q=FailsToDescend;disposition=DiagnosticNotAction;quantum=GaugeInconsistent;anomaly=LocalGauge
SYM-01-REP-ANOMALY-CONTRIBUTION-v2	None	DerivedWitness	q=NotApplicable;disposition=DiagnosticNotAction;quantum=Conditional;anomaly=LocalGauge
SYM-01-SM-ANOMALY-CANCELLATION-v1	R9-CONSISTENCY (ConstituentOfPackage)	DerivedWitness	q=NotApplicable;disposition=DiagnosticNotAction;quantum=Exact;anomaly=None
SYM-01-SU2-GLOBAL-ANOMALY-v1	R9-CONSISTENCY (ConstituentOfPackage)	Obstruction;DerivedWitness	q=FailsToDescend;disposition=DiagnosticNotAction;quantum=GaugeInconsistent;anomaly=GlobalGauge
SYM-01-GLOBAL-FORM-PLACEHOLDER-v1	R9-GLOBALFORM-TPOLOGY (ConstituentOfPackage)	Anchor;TargetUpdate	q=NotApplicable;disposition=StructuralDatum;quantum=Unresolved;anomaly=None
SYM-01-WILSON-LOOP-v1	R9-GLOBALFORM-TPOLOGY (ConstituentOfPackage)	PhysicalTensor;DerivedWitness	q=Nontrivial;disposition=PhysicalTensorDatum;quantum=Conditional;anomaly=None
SYM-01-COUPPLING-NONCLAIM-v1	None	TargetUpdate	q=NotApplicable;disposition=TargetUpdateDisposition;quantum=Conditional;anomaly=None
SYM-01-EW-VACUUM-HANDOFF-v1	None	TargetUpdate	q=NotApplicable;disposition=TargetUpdateDisposition;quantum=Exact;anomaly=None
SYM-01-LINE-OPERATOR-HANDOFF-v1	None	TargetUpdate	q=NotApplicable;disposition=TargetUpdateDisposition;quantum=Unresolved;anomaly=None

Canonical example. The compact-support proper gauge operation occupies the proper-operation slot, is ATS skin because it remains inside the proper physical quotient class, and has quotient-trivial SYM-0 status. Presentation, transport, reduction, response, and consistency occupy separate architecture roles. Therefore

$$\mathcal{T}_{\text{proper quotient-trivial}} \not\approx \mathfrak{A}_{\text{gauge is skin}}.$$

16.4 Exact boundary

The theorem is complete relative to the declared integrated SYM-1 target signature. It does not select the faithful global form, close the line-operator or generalized-symmetry sector, solve electroweak vacuum fixation, determine numerical couplings, establish RG closure, or range over arbitrary UV completions or quantum gravity. Those burdens remain assigned to SYM-8 through SYM-11.

Worked Realization Stress Test: Lane A Sharp-Local Construction

17.1 Purpose and dependency boundary

The Lane A companion is a downstream realization stress test, not an upstream premise of kernel necessity or Standard Model gauge-role minimality. Its value for SYM-1 is methodological: it supplies a source-backed chain in which quotient formation, transport, finite-cap synchronization, mixed-channel descent, sharp-local extension, inductive-union compatibility, and cyclicity are kept theoremmatically distinct. The companion explicitly warns against promoting a fixed-cap theorem into a full-union theorem or treating a bounded-state restriction as the inductive-union passage [16].

17.2 The eight-node realization chain

Node	Realization result	Source location	Scope	SYM-1 use
LA-01	Unique flowed continuum state and bounded positive-time base	Sec. I; App. A, Thm. A.18 and Cor. A.19-A.21	Flowed continuum / bounded positive-time target	Anchor and derived state witness; fixes the state used by later realization nodes
LA-02	Exact-dimension quotient identity and coefficient extraction	Sec. II; App. E, Thm. E.9 and E.12	Fixed exact-dimension quotient block	Quotient-content and derived-witness certificates; no role is inferred from basis notation
LA-03	One-shell transport and finite-truncation inverse control	Sec. III; App. B, Thm. B.7-B.8; Cor. B.9	Fixed finite engineering cap	Comparison/transport realization module and finite-cap continuation certificate
LA-04	Finite mixed bounded-family packaging	Sec. IV; App. D, Cor. D.8	Fixed cap and finite bounded family	Compatible-state packaging witness; explicitly not the closure theorem
LA-05	Finite-cap mixed-correlator closure	Sec. IV; App. D, Thm. D.9	Fixed cap, fixed mixed channel, finite family	Descent and replacement-error certificates; supplies the sole finite-cap mixed-channel closure node
LA-06	Finite-cap sharp-local state extension	Sec. V; App. B, Thm. 5.74A, B.12, B.13	One fixed engineering cap	Positive/unital state, OS structure, and local-field realization; no inductive-union claim
LA-07	Inductive-union passage	Sec. VI; App. B, Cor. B.17	Compatible family of finite caps	Qualitative compatibility and continuation certificate; capwise bounds are not silently promoted to uniform full-union bounds
LA-08	Bounded-base cyclicity	Sec. VI; App. B, Thm. B.18	Reconstructed full sharp-local Hilbert space	Derived density/cyclicity witness; no new spectral-gap theorem

17.3 What the case establishes

Proposition 17.1 (Lane A protocol realization). The eight-node chain instantiates the SYM-1 protocol families for state fixation, quotient/descent, transport, synchronization, complete realization at finite cap, inductive-union continuation, and cyclicity. Each node carries a fixed scope and source locator; no later conclusion is read into an earlier node.

Proof. The flowed state and bounded positive-time base fix the state target. The exact-dimension quotient results supply quotient identity and coefficient extraction before finite-cap closure. One-shell transport and finite-truncation inverse control supply the comparison/continuation input. Finite-family packaging and the mixed-correlator theorem supply separate synchronization and descent nodes. The finite-cap extension supplies positive/unital OS structure at one cap. Compatibility on overlaps then yields the qualitative inductive union, and bounded-base cyclicity is a later density export. This is exactly the source order recorded by the companion. $\square$

17.4 Scope discipline demonstrated by the case

The case exposes four scope errors that the general SYM-1 calculus must block:

1. A finite mixed-family packaging result is not yet a mixed-correlator closure theorem.
2. A finite-cap extension is not an inductive-union construction.
3. Qualitative compatibility on an inductive union does not make cap-dependent quantitative bounds uniform over the full union.
4. Bounded-base cyclicity is a derived density statement, not a new gauge symmetry, a new admissibility gate, or a spectral-gap theorem.

These distinctions provide an external stress test of the target, certificate, and role-occupancy discipline. They do not enlarge the fixed Standard Model target; they show how one admitted Yang–Mills realization must be decomposed without scope laundering.

17.5 What the case does not establish

The companion does not derive the AASC kernel, determine the Standard Model global form, prove the general target-role occupancy theorem, or replace the primary QFT certificates used elsewhere in SYM-1. Its theorems are entered only as realization-specific certificates. Any attempt to use Lane A as an upstream generator would reverse the dependency order and is excluded.

Audit Architecture, Reproducibility, and Claim Control

18.1 Controlled research objects

The active release contains the integrated target signature, target role set, role-necessity certificates, parameterized protocol families, generated candidate-locus universe, presentation classes, complete positive and negative eligibility matrix, all 4,608 integrated injective eligible preassignments, computed compatibility records, derived live residue, complete assignments, lawful automorphism groups, assignment orbits, seven-part completeness certificates, ATS/SYM-0 overlays, source claims, and evidence packets. The manuscript is a rendered projection of those controlled objects.

Controlled object	Count
Integrated target roles	11
Candidate-locus records	75
Presentation classes	70
Eligibility pairs	752
Positive eligibility edges	28
Integrated eligible preassignments	4608
Integrated compatible preassignments	1
Integrated eligible residue	24
Integrated live residue	11
Integrated complete assignments	1
Integrated assignment orbits	1
Chart assignments	2
Chart automorphism group order	4
Chart assignment orbits	1
Hostile tests passed	18/18

18.2 What a PASS establishes

A release PASS establishes the finite result relative to the declared certificate: role positions are fixed before candidate matching; every target-signature candidate is presentation-classified; every residue-role pair has one disposition; every eligible injective assignment has a compatibility calculation; live residue is the union of compatible images; every complete assignment and lawful automorphism has been enumerated; all seven completeness clauses pass; and the assignment quotient has one orbit. It does not convert an imported continuum-QFT theorem into a local proof or establish completeness beyond the declared target signature.

18.3 Executable finite proof

The script `tools/build_v9_integrated_closure.py` regenerates the certificate in the required dependency order. The audit script independently checks counts, matrix totality, compatibility enumeration, live-residue derivation, bijection conditions, group elements, orbit closure, completeness clauses, and hostile tests. Generated outputs are not authoritative inputs to the generator.

18.4 Formalization boundary

The Lean companion specifies finite role-occupancy objects and the imported Role-Occupancy Closure interface. The executable Python model checker is the verified finite calculation in this release environment. Kernel, fixed-domain, ATS, SYM-0, role-occupancy, and continuum-QFT results remain explicit imported theorem interfaces. The release does not claim end-to-end machine proof of interacting quantum gauge theory.

18.5 Reproducible build

The build regenerates v9 data and manuscript tables, runs the integrated-closure audit and project audit, compiles the manuscript with Biber, performs PDF preflight, extracts text and font data, renders every page in PDFium and Poppler, verifies selected-page parity, and writes the release manifest and checksums.

Conclusion

The word *gauge* does not designate one role. A proper compact-support operation, boundary action, large transformation, connection representative, physical holonomy, Gauss-law generator, charge value, BRST cohomology, and gauge anomaly do different work. Their shared notation does not license a shared physical or structural status.

The proper-bulk result is precise: compact-support or boundary-identity gauge operations act within physical quotient classes and are therefore ATS skin at that locus. Their quotient triviality does not remove coherent presentation, comparison and transport, physical reduction, matter response, or quantum consistency. Boundary, global-form/topology, theta, and BRST interfaces are separately typed inside one integrated target rather than closed through isolated microtargets.

The closure calculation is likewise precise. Eleven occupant-independent roles are fixed before residue inspection. Seventy-five generated candidate-locus records form seventy presentation classes. The integrated eligibility matrix contains genuine competition, producing 4,608 injective pointwise eligible preassignments. Every assignment is evaluated against computed transport, reduction, representation, anomaly, proper-operation, boundary, topology, theta, BRST, synchronization, and cross-interface conditions. One integrated preassignment is compatible; its eleven occupants constitute the live residue and one complete bijection. The lawful assignment quotient is singleton. A separate two-chart target has two assignments and a four-element lawful automorphism group, yet still one orbit, demonstrating the nontrivial operation of the quotient.

ATS and SYM-0 remain independent overlays. Diagnostic witnesses, anomaly contributions, charge values, boundary-status records, obstruction certificates, and downstream handoffs are not promoted into architectural roles. This separation allows the paper to state, without contradiction, that proper operations are skin, boundary operations may act physically, BRST is a target-relative realization, anomalies are obstructions, and gauge architecture remains load-bearing.

Accordingly, SYM-1 reaches an endpoint stronger than its original eight-theorem skeleton: the full architecture/operation separation package is supplemented by integrated target-role occupancy closure, a target-derived candidate protocol, complete eligibility and compatibility calculations, derived live residue, an explicit lawful automorphism quotient, and a same-target falsification interface. The result remains narrower than SYM-11. A same-target missing role, candidate, edge, compatibility condition, invariant, or second assignment orbit reopens the theorem.

Final theorem in one sentence

Relative to the frozen integrated SYM-1 certificate, the Standard Model gauge target has one complete compatible target-role occupation modulo only full-invariant-preserving lawful equivalence; proper compact-support gauge operations remain representative skin at the

proper-bulk locus while the integrated gauge architecture remains load-bearing.

Canonical Atlas and Three-Axis Schema

A.1 Schema purpose

The atlas prevents one familiar gauge word from carrying several physically different claims. Every row has a stable identifier, target, candidate kind, locus, action or structure fields, quotient, generator/charge/witness decomposition, quantum and anomaly status, boundary profile, source authority, protocol-family provenance, and completion status.

A.2 Core field families

Field family	Audit question
Identity and target	What immutable object is classified, and in exactly which target, boundary, topology, operator category, quantum scope, and global-form parameter?
Candidate and locus	What structure, action, generator, quotient, bridge, transport, constraint, diagnostic, cohomological object, or target datum is present, and where?
Action decomposition	What transformation, generator, charge value, and witness are separately present?
Invariant and quotient	What is preserved, what equivalence is used, and how does the candidate act after quotienting?
Physical status	What are the classical, quantum, anomaly, vacuum, scale, boundary, and global-form statuses?
Protocol provenance	Which target-derived family generated the candidate-locus unit, and is it an atlas instance, composite package constituent, or external protocol object?
Target-role occupancy	Does the row directly occupy an integrated role, contribute to an occupied composite package, or remain peripheral?
ATS/SYM-0 overlay	What are the independent anchor/tensor/skin type and symmetry-status vector?
Version control	Which earlier row or theorem is superseded, and what changed?

A.3 Active atlas

Record	Object	Target	Constructor	Quotient/action/status	Recorded roles/status
SYM-01-GOVERNANCE-QUOTIENT-v2	Gauge governance/standing quotient	D-SYM1-SM-GOVERNANCE	QuotientBridge	Q-GOV-STANDING;NotApplicable;NotApplicable;Complete	CarrierConstraint
SYM-01-QUOTIENT-FIDELITY-BRIDGE-v2	Governance-to-classical quotient fidelity bridge	D-SYM1-SM-BULK-PROPER	QuotientBridge	Q-GOV-STANDING;NotApplicable;NotApplicable;Complete	DerivedWitness
SYM-01-CHARGE-NORMALIZATION-v1	Overall Abelian charge normalization convention	D-SYM1-SM-ASSEMBLY	Charge	Q-NONE;Trivial;Exact;Complete	Skin
SYM-01-CHARGE-WITNESS-v1	Gauge-charge response witness	D-SYM1-SM-ASSEMBLY	Charge	Q-CL-PROPER-CONFIG;Nontrivial;Exact;Complete	PhysicalTensor;DerivedWitness
SYM-01-CONNECTION-ROLE-v1	Abstract standing-preserving transport-comparison role	D-SYM1-SM-ASSEMBLY	Transport	Q-GOV-STANDING;NotApplicable;Exact;Complete	CarrierConstraint
SYM-01-CURVATURE-ROLE-v1	Curvature as transport-holonomy obstruction role	D-SYM1-SM-ASSEMBLY	LoopResidue	Q-NONE;NotApplicable;Exact;Complete	CarrierConstraint;DerivedWitness
SYM-01-GAUGE-ARCH-v1	Formulation-independent core Standard Model gauge architecture	D-SYM1-SM-ASSEMBLY	Target	Q-GOV-STANDING;NotApplicable;Exact;Complete	CarrierConstraint
SYM-01-HOLONOMY-CLASS-v1	Holonomy or conjugacy-class loop residue	D-SYM1-SM-ASSEMBLY	LoopResidue	Q-GLOBALFORM-DEFERRED;Nontrivial;Exact;Complete	PhysicalTensor;DerivedWitness
SYM-01-LIE-SPECIALIZATION-v1	Lie-type faithful gauge realization criterion	D-SYM1-SM-ASSEMBLY	ProductSector	Q-CL-PROPER-CONFIG;NotApplicable;Conditional;Complete	DerivedWitness
SYM-01-PRODUCT-COMPATIBILITY-v1	Product gauge-sector compatibility ledger	D-SYM1-SM-ASSEMBLY	ProductSector	Q-NONE;NotApplicable;Exact;Complete	CarrierConstraint
SYM-01-REPRESENTATION-RESPONSE-v1	Individual matter representation response	D-SYM1-SM-ASSEMBLY	Representation	Q-CL-PROPER-CONFIG;NotApplicable;Conditional;Complete	CarrierConstraint
SYM-01-SM-ASSEMBLY-CERT-v1	Core Standard Model gauge-architecture assembly certificate	D-SYM1-SM-ASSEMBLY	Target	Q-GOV-STANDING;NotApplicable;Exact;Complete	DerivedWitness
SYM-01-SM-LOCAL-PRODUCT-v1	Local Standard Model gauge product $SU(3)_C \times SU(2)_L \times U(1)_Y$	D-SYM1-SM-ASSEMBLY	ProductSector	Q-NONE;NotApplicable;Exact;Complete	CarrierConstraint
SYM-01-TRANSITION-GROUPOID-v1	Boundary-fixed local/restriction gauge transition groupoid	D-SYM1-SM-ASSEMBLY	Transition	Q-NONE;NotApplicable;Exact;Complete	CarrierConstraint
SYM-01-GAUGE-NATURAL-EQUIVALENCE-v2	Gauge transformation as natural equivalence of transport functors	D-SYM1-SM-BULK-PROPER	Transport	Q-CL-PROPER-CONFIG;Trivial;NotApplicable;Complete	Skin
SYM-01-THIN-PATH-GROUPOID-v2	Thin path groupoid of the base manifold	D-SYM1-SM-ASSEMBLY	Transition	Q-NONE;NotApplicable;NotApplicable;Complete	CarrierConstraint
SYM-01-TRANSPORT-FUNCTOR-v2	Smooth parallel-transport functor	D-SYM1-SM-ASSEMBLY	Transport	Q-CL-PROPER-CONFIG;Mixed;NotApplicable;Complete	CarrierConstraint
SYM-01-CLASSICAL-GAUGE-QUOTIENT-v2	Classical proper-gauge configuration quotient	D-SYM1-SM-BULK-PROPER	QuotientBridge	Q-CL-PROPER-CONFIG;NotApplicable;NotApplicable;Complete	CarrierConstraint
SYM-01-CONNECTION-DYNAMICAL-VARIATION-v2	Transverse/dynamical variation of a connection class	D-SYM1-SM-BULK-PROPER	Transport	Q-CL-PROPER-CONFIG;Nontrivial;Conditional;Complete	PhysicalTensor
SYM-01-CONNECTION-LOCAL-COORDINATES-v2	Local connection one-form coordinates A	D-SYM1-SM-BULK-PROPER	Transport	Q-CL-PROPER-CONFIG;Trivial;NotApplicable;Complete	Skin
SYM-01-CONNECTION-ORBIT-CLASS-v2	Physical connection configuration class $[A]$	D-SYM1-SM-BULK-PROPER	Transport	Q-CL-PROPER-CONFIG;Nontrivial;Conditional;Complete	PhysicalTensor
SYM-01-CONNECTION-VERTICAL-VARIATION-v2	Vertical gauge variation of a connection representative	D-SYM1-SM-BULK-PROPER	Operation	Q-CL-PROPER-CONFIG;Trivial;Conditional;Complete	Skin
SYM-01-CONSTRAINT-SURFACE-v1	Gauge-theory standing/constraint surface	D-SYM1-SM-BULK-PROPER	Constraint	Q-NONE;NotApplicable;Exact;Complete	CarrierConstraint
SYM-01-CURVATURE-FORM-v1	Local curvature form $F = dA + A \wedge A$	D-SYM1-SM-BULK-PROPER	LoopResidue	Q-CL-PROPER-CONFIG;Mixed;Exact;Complete	PhysicalTensor;DerivedWitness
SYM-01-GAUGE-LOCAL-PROPER-v1	Compact-support proper local gauge operation	D-SYM1-SM-BULK-PROPER	Operation	Q-CL-PROPER-CONFIG;Trivial;Exact;Complete	Skin
SYM-01-GAUSS-GENERATOR-PROPER-v1	Proper Gauss-law generator	D-SYM1-SM-BULK-PROPER	Constraint	Q-REDUCED-PHASE;Trivial;Exact;Complete	DerivedWitness
SYM-01-REPRESENTATION-BASIS-v1	Representation basis change	D-SYM1-SM-BULK-PROPER	Representation	Q-NONE;Trivial;Exact;Complete	Skin
SYM-01-GAUGE-BOUNDARY-ACTION-v1	Boundary or asymptotic gauge transformation with nonzero charge	D-SYM1-SM-BOUNDARY	Operation	Q-BOUNDARY-MIXED;Mixed;Conditional;Complete	PhysicalAction
SYM-01-GAUGE-BOUNDARY-RULE-v1	Boundary gauge-action product-status rule	D-SYM1-SM-BOUNDARY	Operation	Q-BOUNDARY-MIXED;Mixed;Conditional;Complete	DerivedWitness
SYM-01-SURFACE-CHARGE-v1	Boundary surface charge	D-SYM1-SM-BOUNDARY	Charge	Q-BOUNDARY-MIXED;Nontrivial;Exact;Complete	PhysicalTensor;DerivedWitness
SYM-01-GAUGE-LARGE-EQUIVARIANCE-v2	Large-gauge wavefunctional equivariance phase	D-SYM1-SM-THETA	Operation	Q-PROJECTIVE-RAY;Trivial;Conditional;Complete	DerivedWitness

Record	Object	Target	Constructor	Quotient/action/status	Recorded roles/status
SYM-01-GAUGE-LARGE-QUOTIENT-v1	Large gauge transformation declared in the physical quotient	D-SYM1-SM-TOP0-QUOTIENT	QuotientBridge	Q-LARGE-DECLARED;Trivial;Exact;Comp	Skin
SYM-01-GAUGE-LARGE-SECTOR-ACTION-RULE-v2	Conditional physical action of a large transformation	D-SYM1-SM-THETA	Operation	Q-PROJECTIVE-RAY;Mixed;Conditional;Complete	PhysicalAction
SYM-01-THETA-SECTOR-DATUM-v2	Theta-sector or quantization character datum	D-SYM1-SM-THETA	Operation	Q-PROJECTIVE-RAY;NotApplicable;Conditional;Complete	Anchor
SYM-01-TOPOLOGY-ANCHOR-v1	Fixed bundle and topological-sector datum	D-SYM1-SM-THETA	Operation	Q-LARGE-DECLARED;NotApplicable;Exact;Complete	Anchor
SYM-01-TOPOLOGY-UPDATE-v1	Admission of a new gauge-topological sector	D-SYM1-SM-TOP0-QUOTIENT	Operation	Q-LARGE-DECLARED;NotApplicable;Exact;Complete	TargetUpdate
SYM-01-BRST-DIFFERENTIAL-v1	Nilpotent BRST differential	D-SYM1-SM-BRST	BRST	Q-BRST-LOCAL;Mixed;PerturbativelyExact;Complete	DerivedWitness
SYM-01-BRST-REALIZATION-ARCH-v2	Target-relative BRST realization architecture	D-SYM1-SM-BRST	BRST	Q-BRST-LOCAL;NotApplicable;PerturbativelyExact;Complete	CarrierConstraint
SYM-01-FP-GHOSTS-v1	Faddeev-Popov ghost and antighost fields	D-SYM1-SM-BRST	GaugeFixing	Q-BRST-LOCAL;Trivial;PerturbativelyExact;Complete	Skin;DerivedWitness
SYM-01-GAUGE-FIXING-v1	Gauge-fixing condition or gauge-fixing fermion	D-SYM1-SM-BRST	GaugeFixing	Q-BRST-LOCAL;Trivial;PerturbativelyExact;Complete	Skin
SYM-01-GHOST-NUMBER-v1	Ghost-number grading	D-SYM1-SM-BRST	BRST	Q-BRST-LOCAL;NotApplicable;Conditional;Complete	DerivedWitness
SYM-01-GRIBOV-OBSTRUCTION-v1	Gribov/Singer obstruction to global gauge fixing	D-SYM1-GRIBOV-TEST	GaugeFixing	Q-CL-PROPER-CONFIG;Nontrivial;Conditional;Complete	Obstruction;DerivedWitness
SYM-01-BRST-STATE-COHOMOLOGY-v2	BRST charge cohomology of physical states	D-SYM1-SM-BRST-STATE	BRST	Q-BRST-STATE;Nontrivial;PerturbativelyExact;Complete	DerivedWitness
SYM-01-BRST-LOCAL-COHOMOLOGY-v2	Local BRST cohomology of operators/functions	D-SYM1-SM-BRST-LOCAL	BRST	Q-BRST-LOCAL;Nontrivial;PerturbativelyExact;Complete	DerivedWitness
SYM-01-BRST-RELATIVE-COHOMOLOGY-v2	Relative local BRST cohomology $H^{\bullet}_{\{g,n\}}(s d)$	D-SYM1-SM-BRST-RELATIVE	BRST	Q-BRST-RELATIVE;Nontrivial;PerturbativelyExact;Complete	DerivedWitness
SYM-01-AGGREGATE-REP-LEDGER-v2	Aggregate chiral representation and anomaly ledger	D-SYM1-SM-ANOMALY	Representation	Q-QUANTUM-DESCENT;NotApplicable;Exact;Complete	CarrierConstraint;DerivedWitness
SYM-01-ANOMALY-NEUTRALITY-v1	Aggregate gauge-anomaly neutrality condition	D-SYM1-SM-ANOMALY	QuantumObstruction	Q-QUANTUM-DESCENT;NotApplicable;Exact;Complete	CarrierConstraint;DerivedWitness
SYM-01-INFLOW-UPDATE-v1	Anomaly inflow or added-sector neutralization	D-SYM1-SM-ANOMALY	QuantumObstruction	Q-QUANTUM-DESCENT;NotApplicable;Conditional;Complete	TargetUpdate
SYM-01-LOCAL-GAUGE-ANOMALY-v1	Noncanceled perturbative local gauge anomaly	D-SYM1-SM-ANOMALY	QuantumObstruction	Q-QUANTUM-DESCENT;FailsToDescend;GaugeInconsistent;Complete	Obstruction
SYM-01-REP-ANOMALY-CONTRIBUTION-v2	Individual chiral representation anomaly contribution	D-SYM1-SM-ANOMALY	Representation	Q-QUANTUM-DESCENT;NotApplicable;Conditional;Complete	DerivedWitness
SYM-01-SM-ANOMALY-CANCELLATION-v1	Standard Model perturbative gauge-anomaly cancellation ledger	D-SYM1-SM-ANOMALY	QuantumObstruction	Q-QUANTUM-DESCENT;NotApplicable;Exact;Complete	DerivedWitness
SYM-01-SU2-GLOBAL-ANOMALY-v1	Witten SU(2) global gauge-anomaly obstruction	D-SYM1-SM-ANOMALY	QuantumObstruction	Q-QUANTUM-DESCENT;FailsToDescend;GaugeInconsistent;Complete	Obstruction;DerivedWitness
SYM-01-GLOBAL-FORM-PLACEHOLDER-v1	Faithful Standard Model global gauge-group form G_{SM}/Γ	D-SYM1-SM-GLOBALFORM-PREVIEW	TargetEdge	Q-GLOBALFORM-DEFERRED;NotApplicable;Unresolved;Deferred	Anchor;TargetUpdate
SYM-01-WILSON-LOOP-v1	Traced Wilson-loop observable	D-SYM1-SM-GLOBALFORM-PREVIEW	LoopResidue	Q-GLOBALFORM-DEFERRED;Nontrivial;Conditional;Complete	PhysicalTensor;DerivedWitness
SYM-01-COUPPING-NONCLAIM-v1	Gauge-coupling values and running	D-SYM1-SM-ASSEMBLY	TargetEdge	Q-NONE;NotApplicable;Conditional;Deferred	TargetUpdate
SYM-01-EW-VACUUM-HANDOFF-v1	Electroweak vacuum fixation and residual $U(1)_{em}$	D-SYM1-EW-HANDOFF	TargetEdge	Q-NONE;NotApplicable;Exact;Deferred	TargetUpdate
SYM-01-LINE-OPERATOR-HANDOFF-v1	Line-operator and one-form-symmetry completion	D-SYM1-SM-GLOBALFORM-PREVIEW	TargetEdge	Q-GLOBALFORM-DEFERRED;NotApplicable;Unresolved;Deferred	TargetUpdate

A.4 Three-axis classification

Target-role occupancy, ATS type, and SYM-0 status are separately stored. Multi-tagging is meaningful only across distinct questions or loci.

Record	Target-role occupancy	ATS	SYM-0 status
SYM-01-GOVERNANCE-QUOTIENT-v2	None	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=NotApplicable;anomaly=NotApplicable
SYM-01-QUOTIENT-FIDELITY-BRIDGE-v2	R9-REDUCTION(ConstituentOfPackage)	DerivedWitness	q=NotApplicable;disposition=QuotientBridge;quantum=NotApplicable;anomaly=None
SYM-01-CHARGE-NORMALIZATION-v1	None	Skin	q=Trivial;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-CHARGE-WITNESS-v1	None	PhysicalTensor;DerivedWitness	q=Nontrivial;disposition=PhysicalTensorDatum;quantum=Exact;anomaly=None
SYM-01-CONNECTION-ROLE-v1	None	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-CURVATURE-ROLE-v1	None	CarrierConstraint;DerivedWitness	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-GAUGE-ARCH-v1	R9-SYNCHRONIZATION(ConstituentOfPackage)	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-HOLONOMY-CLASS-v1	None	PhysicalTensor;DerivedWitness	q=Nontrivial;disposition=PhysicalTensorDatum;quantum=Exact;anomaly=None
SYM-01-LIE-SPECIALIZATION-v1	R9-SYNCHRONIZATION(ConstituentOfPackage)	DerivedWitness	q=NotApplicable;disposition=StructuralDatum;quantum=Conditional;anomaly=None
SYM-01-PRODUCT-COMPATIBILITY-v1	R9-SYNCHRONIZATION(ConstituentOfPackage)	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-REPRESENTATION-RESPONSE-v1	R9-RESPONSE(DirectOccupant)	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=Conditional;anomaly=None
SYM-01-SM-ASSEMBLY-CERT-v1	R9-SYNCHRONIZATION(ConstituentOfPackage)	DerivedWitness	q=NotApplicable;disposition=DiagnosticNotAction;quantum=Exact;anomaly=None
SYM-01-SM-LOCAL-PRODUCT-v1	R9-PRESENTATION(ConstituentOfPackage)	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-TRANSITION-GROUPOID-v1	R9-PRESENTATION(ConstituentOfPackage)	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-GAUGE-NATURAL-EQUIVALENCE-v2	None	Skin	q=Trivial;disposition=TrivialAction;quantum=NotApplicable;anomaly=None
SYM-01-THIN-PATH-GROUPOID-v2	None	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=NotApplicable;anomaly=None
SYM-01-TRANSPORT-FUNCTOR-v2	R9-COMPARISON(DirectOccupant)	CarrierConstraint	q=Mixed;disposition=StructuralDatum;quantum=NotApplicable;anomaly=None
SYM-01-CLASSICAL-GAUGE-QUOTIENT-v2	R9-REDUCTION(ConstituentOfPackage)	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=NotApplicable;anomaly=None
SYM-01-CONNECTION-DYNAMICAL-VARIATION-v2	None	PhysicalTensor	q=Nontrivial;disposition=PhysicalTensorDatum;quantum=Conditional;anomaly=None
SYM-01-CONNECTION-LOCAL-COORDINATES-v2	None	Skin	q=Trivial;disposition=StructuralDatum;quantum=NotApplicable;anomaly=None
SYM-01-CONNECTION-ORBIT-CLASS-v2	None	PhysicalTensor	q=Nontrivial;disposition=PhysicalTensorDatum;quantum=Conditional;anomaly=None
SYM-01-CONNECTION-VERTICAL-VARIATION-v2	None	Skin	q=Trivial;disposition=TrivialAction;quantum=Conditional;anomaly=None
SYM-01-CONSTRAINT-SURFACE-v1	R9-REDUCTION(ConstituentOfPackage)	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-CURVATURE-FORM-v1	None	PhysicalTensor;DerivedWitness	q=Mixed;disposition=PhysicalTensorDatum;quantum=Exact;anomaly=None
SYM-01-GAUGE-LOCAL-PROPER-v1	R9-PROPER-OPERATION(DirectOccupant)	Skin	q=Trivial;disposition=TrivialAction;quantum=Exact;anomaly=None
SYM-01-GAUSS-GENERATOR-PROPER-v1	R9-REDUCTION(ConstituentOfPackage)	DerivedWitness	q=Trivial;disposition=GeneratorNotAction;quantum=Exact;anomaly=None
SYM-01-REPRESENTATION-BASIS-v1	None	Skin	q=Trivial;disposition=TrivialAction;quantum=Exact;anomaly=None
SYM-01-GAUGE-BOUNDARY-ACTION-v1	R9-BOUNDARY-ACTION(ConstituentOfPackage)	PhysicalAction	q=Mixed;disposition=NontrivialAction;quantum=Conditional;anomaly=None
SYM-01-GAUGE-BOUNDARY-RULE-v1	R9-BOUNDARY-ACTION(ConstituentOfPackage)	DerivedWitness	q=Mixed;disposition=DiagnosticNotAction;quantum=Conditional;anomaly=None
SYM-01-SURFACE-CHARGE-v1	R9-BOUNDARY-ACTION(ConstituentOfPackage)	PhysicalTensor;DerivedWitness	q=Nontrivial;disposition=PhysicalTensorDatum;quantum=Exact;anomaly=None

Record	Target-role occupancy	ATS	SYM-0 status
SYM-01-GAUGE-LARGE-EQUIVARIANCE-v2	R9-THETA-REALIZATION(ConstituentOfPackage)	DerivedWitness	q=Trivial;disposition=DiagnosticNotAction;quantum=Conditional;anomaly=None
SYM-01-GAUGE-LARGE-QUOTIENT-v1	None	Skin	q=Trivial;disposition=TrivialAction;quantum=Exact;anomaly=None
SYM-01-GAUGE-LARGE-SECTOR-ACTION-RULE-v2	R9-THETA-REALIZATION(ConstituentOfPackage)	PhysicalAction	q=Mixed;disposition=ConditionalActionRule;quantum=Conditional;anomaly=None
SYM-01-THETA-SECTOR-DATUM-v2	R9-THETA-REALIZATION(ConstituentOfPackage)	Anchor	q=NotApplicable;disposition=StructuralDatum;quantum=Conditional;anomaly=None
SYM-01-TOPOLOGY-ANCHOR-v1	R9-GLOBALFORM-TOPOLOGY(ConstituentOfPackage);R9-THETA-REALIZATION(ConstituentOfPackage)	Anchor	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-TOPOLOGY-UPDATE-v1	None	TargetUpdate	q=NotApplicable;disposition=TargetUpdateDisposition;quantum=Exact;anomaly=None
SYM-01-BRST-DIFFERENTIAL-v1	R9-BRST-REALIZATION(ConstituentOfPackage)	DerivedWitness	q=Mixed;disposition=CohomologicalProjection;quantum=PerturbativelyExact;anomaly=BRST
SYM-01-BRST-REALIZATION-ARCH-v2	R9-BRST-REALIZATION(ConstituentOfPackage)	CarrierConstraint	q=NotApplicable;disposition=StructuralDatum;quantum=PerturbativelyExact;anomaly=BRST
SYM-01-FP-GHOSTS-v1	R9-BRST-REALIZATION(ConstituentOfPackage)	Skin;DerivedWitness	q=Trivial;disposition=StructuralDatum;quantum=PerturbativelyExact;anomaly=BRST
SYM-01-GAUGE-FIXING-v1	R9-BRST-REALIZATION(ConstituentOfPackage)	Skin	q=Trivial;disposition=StructuralDatum;quantum=PerturbativelyExact;anomaly=None
SYM-01-GHOST-NUMBER-v1	None	DerivedWitness	q=NotApplicable;disposition=DiagnosticNotAction;quantum=Conditional;anomaly=BRST
SYM-01-GRIBOV-OBSTRUCTION-v1	None	Obstruction;DerivedWitness	q=Nontrivial;disposition=DiagnosticNotAction;quantum=Conditional;anomaly=None
SYM-01-BRST-STATE-COHOMOLOGY-v2	R9-BRST-REALIZATION(ConstituentOfPackage)	DerivedWitness	q=Nontrivial;disposition=CohomologicalProjection;quantum=PerturbativelyExact;anomaly=BRST
SYM-01-BRST-LOCAL-COHOMOLOGY-v2	R9-BRST-REALIZATION(ConstituentOfPackage)	DerivedWitness	q=Nontrivial;disposition=CohomologicalProjection;quantum=PerturbativelyExact;anomaly=BRST
SYM-01-BRST-RELATIVE-COHOMOLOGY-v2	R9-BRST-REALIZATION(ConstituentOfPackage)	DerivedWitness	q=Nontrivial;disposition=CohomologicalProjection;quantum=PerturbativelyExact;anomaly=BRST
SYM-01-AGGREGATE-REP-LEDGER-v2	R9-CONSISTENCY(ConstituentOfPackage)	CarrierConstraint;DerivedWitness	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-ANOMALY-NEUTRALITY-v1	R9-CONSISTENCY(ConstituentOfPackage)	CarrierConstraint;DerivedWitness	q=NotApplicable;disposition=StructuralDatum;quantum=Exact;anomaly=None
SYM-01-INFLOW-UPDATE-v1	None	TargetUpdate	q=NotApplicable;disposition=TargetUpdateDisposition;quantum=Conditional;anomaly=Unresolved
SYM-01-LOCAL-GAUGE-ANOMALY-v1	None	Obstruction	q=FailsToDescend;disposition=DiagnosticNotAction;quantum=GaugeInconsistent;anomaly=LocalGauge
SYM-01-REP-ANOMALY-CONTRIBUTION-v2	None	DerivedWitness	q=NotApplicable;disposition=DiagnosticNotAction;quantum=Conditional;anomaly=LocalGauge
SYM-01-SM-ANOMALY-CANCELLATION-v1	R9-CONSISTENCY(ConstituentOfPackage)	DerivedWitness	q=NotApplicable;disposition=DiagnosticNotAction;quantum=Exact;anomaly=None
SYM-01-SU2-GLOBAL-ANOMALY-v1	R9-CONSISTENCY(ConstituentOfPackage)	Obstruction;DerivedWitness	q=FailsToDescend;disposition=DiagnosticNotAction;quantum=GaugeInconsistent;anomaly=GlobalGauge
SYM-01-GLOBAL-FORM-PLACEHOLDER-v1	R9-GLOBALFORM-TOPOLOGY(ConstituentOfPackage)	Anchor;TargetUpdate	q=NotApplicable;disposition=StructuralDatum;quantum=Unresolved;anomaly=None
SYM-01-WILSON-LOOP-v1	R9-GLOBALFORM-TOPOLOGY(ConstituentOfPackage)	PhysicalTensor;DerivedWitness	q=Nontrivial;disposition=PhysicalTensorDatum;quantum=Conditional;anomaly=None
SYM-01-COUPLING-NONCLAIM-v1	None	TargetUpdate	q=NotApplicable;disposition=TargetUpdateDisposition;quantum=Conditional;anomaly=None
SYM-01-EW-VACUUM-HANDOFF-v1	None	TargetUpdate	q=NotApplicable;disposition=TargetUpdateDisposition;quantum=Exact;anomaly=None
SYM-01-LINE-OPERATOR-HANDOFF-v1	None	TargetUpdate	q=NotApplicable;disposition=TargetUpdateDisposition;quantum=Unresolved;anomaly=None

Target Registry

The target registry is canonical. A change of boundary, topology, global form, operator category, BRST complex, anomaly-neutralizing sector, or vacuum is not silently transported across rows.

Target	Name	Scope discriminators	Purpose
D-SYM1-SM-ASSEMBLY	Integrated Standard Model gauge target	Boundary: Locus-indexed proper-bulk and declared boundary/asymptotic strata; topology: Fixed sharp-local local sector with admitted G_{Γ} family and θ /topology preview; operators: Local fields, physical states and observables, proper/boundary/topological operations, and declared BRST realization interfaces; line-operator completion deferred; quantum: Anomaly-free Standard Model quantum gauge architecture with declared Hamiltonian and BRST realization interfaces	Primary integrated SYM-1 closure target: coherent presentation, comparison/transport, physical reduction, matter response, quantum consistency, proper and boundary operations, topology/-global form, θ realization, BRST realization, and cross-realization synchronization.
D-SYM1-SM-BULK-PROPER	Proper compact-support bulk gauge target	Boundary: Gauge parameter compactly supported or zero at the admitted boundary; topology: Fixed trivial or declared local sector; operators: Gauge-invariant local observables and physical states; quantum: Anomaly-free gauge theory	Small-gauge quotient theorem and Gauss-law classification
D-SYM1-SM-BOUNDARY	Boundary/asymptotic gauge-action target	Boundary: Nonvanishing admitted boundary/asymptotic gauge parameter; differentiable generator required; topology: Fixed; operators: Bulk observables, boundary data, surface charges, possible edge sectors; quantum: Classical/canonical and quantum Ward-identity compatible	Boundary exception theorem and mixed quotient action
D-SYM1-SM-TOPO-QUOTIENT	Large transformation included in quotient target	Boundary: Fixed; topology: Large gauge transformations declared quotient identifications; operators: States and observables invariant under declared large quotient; quantum: Conditional on θ representation and global form	Large-gauge quotient record
D-SYM1-SM-THETA	Theta-sector representation target	Boundary: Fixed; topology: Large transformations represented nontrivially on θ sectors; operators: States, partition function, topological observables; quantum: Quantum topological sector	Large-gauge/topology physical-action record
D-SYM1-SM-BRST	Gauge-fixed perturbative BRST target	Boundary: Boundary conditions compatible with BRST charge; topology: Perturbative local sector; operators: Gauge-fixed fields and BRST cohomology classes; quantum: Perturbative anomaly-free renormalized gauge theory	Parent gauge-fixed perturbative target; state, local-operator and relative local cohomologies are child targets.

Target	Name	Scope discriminators	Purpose
D-SYM1-SM-ANOMALY	Chiral gauge-anomaly test target	Boundary: Fixed; topology: Local and selected global anomaly probes separated; operators: Effective action, measure, currents, partition function; quantum: Regularized/renormalized quantum theory	Gauge-anomaly admissibility and aggregate cancellation
D-SYM1-SM-GLOBALFORM-PREVIEW	Faithful global-form preview target	Boundary: Fixed; topology: Global group G_{SM}/Γ left as explicit target datum; operators: Local operators; line spectrum not closed in SYM-1; quantum: Quantum gauge theory	Parameterize global-sensitive constructions by G_{SM}/Γ while deferring line-operator and generalized-symmetry closure to SYM-9.
D-SYM1-GRIBOV-TEST	Global gauge-fixing obstruction test	Boundary: Fixed; topology: Configuration-space topology admitted; operators: Gauge orbits and gauge-fixing slices; quantum: Classical configuration space and perturbative implications	Classify Gribov/Singer obstruction and block canonical representative privilege
D-SYM1-EW-HANDOFF	Electroweak vacuum-fixation handoff target	Boundary: Fixed; topology: Fixed; operators: Gauge-invariant spectrum and residual charge sectors; quantum: Perturbative electroweak theory	Declare SYM-8 handoff; no local-gauge ontic breaking theorem in SYM-1
D-SYM1-SM-GOVERNANCE	Gauge governance/standing-quotient target	Boundary: Inherited from parent target and held fixed; topology: Inherited from parent target; operators: Standing-bearing gauge claims and their admitted tests; quantum: Neutral with respect to classical versus quantum realization; no physical quotient identified by stipulation	Define the governance quotient and prove target-specific comparison bridges before importing factorization to a physical gauge quotient.
D-SYM1-SM-BRST-STATE	BRST state-cohomology target	Boundary: Boundary conditions compatible with the selected cohomological complex; topology: Perturbative/local sector unless explicitly declared; operators: Extended state complex and BRST charge; quantum: Perturbative anomaly-free BRST realization with nilpotent differential/charge	Separate physical-state cohomology from local operator cohomology.
D-SYM1-SM-BRST-LOCAL	BRST local-operator cohomology target	Boundary: Boundary conditions compatible with the selected cohomological complex; topology: Perturbative/local sector unless explicitly declared; operators: Local fields, jets, ghosts and antifields; quantum: Perturbative anomaly-free BRST realization with nilpotent differential/charge	Classify local BRST cohomology $H^*(g(s))$ without conflating it with states.
D-SYM1-SM-BRST-RELATIVE	BRST relative local-cohomology target	Boundary: Boundary conditions compatible with the selected cohomological complex; topology: Perturbative/local sector unless explicitly declared; operators: Local differential forms modulo spacetime exterior derivative; quantum: Perturbative anomaly-free BRST realization with nilpotent differential/charge	Classify $H^*(g,n)(s d)$, integrated densities, deformations and anomaly candidates.
D-SYM1-TWO-CHART-PRESENTATION	Two-chart interchangeable presentation target	Boundary: No independent boundary charge; topology: Fixed trivial local bundle sector; operators: Local presentations and overlap maps; quantum: Classical presentation test	Controlled nontrivial certificate-lawful automorphism test: two chart roles and two presentation residues form two assignments in one S^2 orbit.

B.1 Target comparison rule

A proposed counterexample must state whether it preserves every target field. If not, it is evaluated as a target update. This rule does not dismiss the proposal; it places it in the correct

comparison class and records which prior results transport.

B.2 Deferred targets

Four active records are explicit handoffs: faithful global form and line operators to SYM-9, electroweak vacuum fixation to SYM-8, and numerical coupling/RG data to SYM-10. They remain in the atlas so that omission cannot be mistaken for closure.

Theorem Inventory

The inventory records status and source burden independently of prose numbering. “Mechanically audited” means a finite data-agreement check, not machine proof of an imported QFT theorem.

ID	Title	Status	Statement	Sources
SYM1-T0	Kernel inheritance for determinate gauge claims	Imported and specialized	Every determinate same-domain gauge claim instantiates Admissibility, Standing, Reference and Irreversibility.	MaleyKernel;MaleyFixedDomain
SYM1-T1	Gauge target-family fixation	Locally proved	Bulk, boundary, topological, BRST-state, BRST-local, BRST-relative, anomaly and global-form targets are distinct and target changes are explicit.	MaleyArc;MaleySYM0
SYM1-T2	Quotient-family separation	Locally proved definition theorem	Governance, classical configuration, reduced phase-space, projective-ray, BRST-state, BRST-local, BRST-relative and quantum-descent quotients are distinct typed objects.	MaleyFixedDomain;HenneauxTeitelboim1992;BarnichBrandtHenneaux2000
SYM1-T3	Proper-bulk quotient fidelity bridge	Locally proved conditional	Gauge-invariant separating admitted tests induce a conditional comparison/isomorphism between governance standing classes and proper classical gauge orbits.	MaleyFixedDomain;MaleySYM0;HenneauxTeitelboim1992
SYM1-T4	BRST quotient comparison family	Imported and specialized conditional	State, local-operator and relative local BRST cohomologies receive separate comparison bridges; none is identified with the others by notation.	KugoOjima1979;BarnichBrandtHenneaux2000;HenneauxTeitelboim1992
SYM1-T5	Gauge architecture/operation separation	Locally proved composition	Load-bearing core architecture is not identical to individual proper local gauge operations.	MaleyR8Gauge;MaleyAER;HenneauxTeitelboim1992
SYM1-T6	Boundary-fixed transition groupoid	Imported	Admitted local gauge transitions minimally form a boundary-fixed local/restriction groupoid.	MaleyA1Groupoid
SYM1-T7	Transition/path/transport separation	Locally proved specialization	Cech transition arrows, thin paths, transport functors and gauge-natural equivalences are distinct typed layers.	SchreiberWaldorf2009;MaleyA2Connection
SYM1-T8	Standing-preserving transport realization	Imported and specialized conditional	Stable endpoint comparison requires transport structure; a smooth transport functor faithfully realizes it under principal-bundle/descent hypotheses.	MaleyA2Connection;SchreiberWaldorf2009
SYM1-T9	Curvature/holonomy role split	Imported and specialized	Curvature and holonomy are physical tensor/observable content and witnesses; they are not symmetry actions.	MaleyA3Curvature;Nakahara2003;Wilson1974
SYM1-T10	Lie-type faithful realization	Conditional realization theorem	Given principal G-bundle/local triviality and smooth transport, A and F faithfully realize antecedent roles; no bare groupoid-to-fixed-G derivation is claimed.	MaleyPSM1;SchreiberWaldorf2009;Nakahara2003
SYM1-T11	Product-sector compatibility	Imported	Product notation is licensed only after co-scope, quotient, transition, transport and mixed-residue tests.	MaleyPSM2
SYM1-T12	Individual representation response	Imported and specialized	An individual representation can be classically admissible while contributing a nonzero anomaly coefficient.	MaleyA4RepCharge;Bilal2008
SYM1-T13	Aggregate quantum representation ledger	Imported and specialized	Quantum gauge admissibility is imposed on the complete chiral representation ledger through cancellation of all required obstructions.	MaleyA5Anomaly;Bilal2008;Witten1982
SYM1-T14	Characteristic quotient identification	Imported with QFT certificate	Under regularity, observable separation and maximality, proper gauge directions equal the presymplectic characteristic distribution.	MaleyAER;HenneauxTeitelboim1992
SYM1-T15	Small-gauge quotient theorem	Locally integrated QFT theorem	Proper compact-support transformations preserving boundary data act trivially on the certified classical quotient.	HenneauxTeitelboim1992;Dirac1964
SYM1-T16	Boundary product-status theorem	Locally integrated conditional theorem	Bulk action, boundary action, charge, flux and edge-data statuses form a product profile; overlapping features are permitted and typed.	ReggeTeitelboim1974;HenneauxTeitelboim1992
SYM1-T17	Large-gauge/theta split	Locally proved classification	A wavefunctional phase is trivial on projective rays unless an independent sector/operator/observable certificate establishes physical action.	WeinbergQFT2;AharonySeibergTachikawa2013
SYM1-T18	Gauge-fixing no-authority theorem	Locally proved composition	Gauge fixing is representative machinery and cannot become a second physical quotient or canonical authorizer.	FaddeevPopov1967;Gribov1978;Singer1978;MaleyFixedDomain
SYM1-T19	BRST three-complex theorem	Imported and specialized	Physical-state, local-operator and relative local BRST cohomologies are distinct and target-indexed.	KugoOjima1979;BarnichBrandtHenneaux2000
SYM1-T20	Gauge-anomaly descent theorem	Imported and specialized	A noncanceled gauge anomaly is failure of quantum descent; factorization is not assumed in the anomalous target.	Bilal2008;BarnichBrandtHenneaux2000;MaleyA5Anomaly
SYM1-T21	Formulation-independent gauge-role core	Locally proved role theorem	Presentation coherence, comparison, physical reduction, matter response, and quantum consistency are the formulation-independent role obligations; smooth, Hamiltonian, BRST/BV, and lattice objects are faithful realizations.	MaleyStandaloneSM;MaleyR8Gauge
SYM1-T22	Target-relative BRST assembly	Locally proved composition	Gauge fixing, ghosts, differential and the three cohomologies are minimal only relative to the declared BRST realization target.	BecchiRouetStora1976;KugoOjima1979;BarnichBrandtHenneaux2000
SYM1-T23	Derivability-sensitive role-core minimality	Locally proved conditional on declared closure rules	No role-core obligation lies in the derivability closure of the remaining obligations; realization-level witnesses are not promoted to primitive roles.	MaleySYM0;MaleyFixedDomain
SYM1-T24	Atomic semantic role uniqueness	Superseded in v4	Every complete atomic semantic effect profile has a unique role at each effect projection, independently of the executable classifier.	MaleySYM0;MaleyFixedDomain
SYM1-T25	Canonical effect-support decomposition	Superseded in v4	Every complete gauge object has a canonical finite multiset of nonzero semantic effect projections.	MaleySYM0;MaleyFixedDomain
SYM1-T26	Lawful-refinement invariance	Superseded in v4	Any admissible decomposition preserving semantic support has a common lawful refinement and the same normalized role multiset.	MaleyFixedDomain;MaleySYM0
SYM1-T27	Executable classifier implementation agreement	Superseded in v4	The answer-excluding executable classifier agrees with the independently defined semantic role normal form for all active records.	MaleySYM0
SYM1-T28	Physical target-language formation completeness	Superseded in v4	Every gauge-relevant proposition formable in the frozen physical signature has a finite derivation; same-target uniformability is a schema-gap counterexample.	MaleyFixedDomain;MaleySYM0

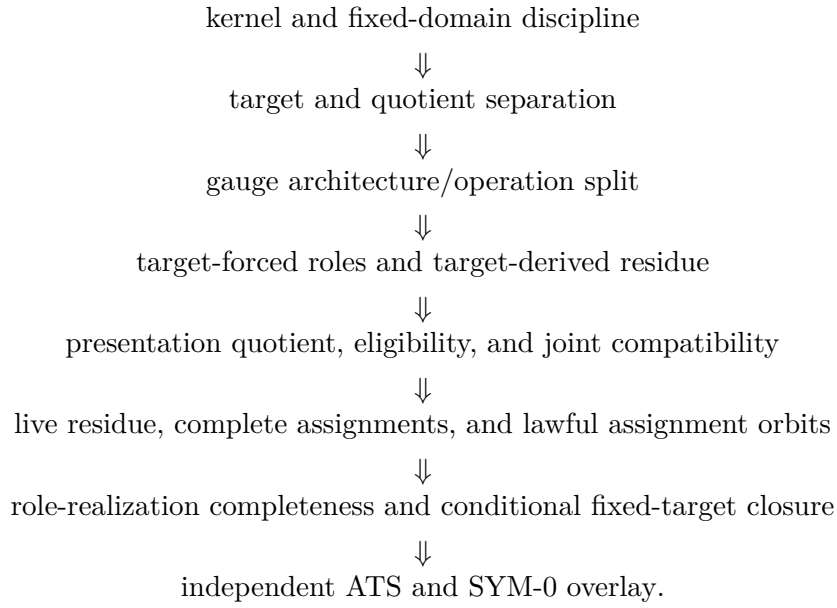
ID	Title	Status	Statement	Sources
SYM1-T29	Composition-specific normalization induction	Superseded in v4	Every well-typed atomic or composite term normalizes by constructor-specific semantic rules for product, aggregation, quotient, descent, cohomology, synchronization, cancellation, boundary restriction, and target extension.	MaleySYM0
SYM1-T30	Schema/target disposition separation	Superseded in v4	Same-target schema extension, physical target update, and outside-subject proposals are distinct dispositions; a same-target schema gap falsifies current adequacy.	MaleySYM0;MaleyFixedDomain
SYM1-T31	Certificate-relative sharp-local normalization exhaustion	Superseded in v4	For every well-typed term in the frozen physical signature, satisfaction of its named QFT certificates guarantees a canonical role normal form; atlas membership and hostile-ledger emptiness are not premises.	MaleyFixedDomain;MaleySYM0
SYM1-T32	Global-form parameterization discipline	Locally proved scope theorem	Every global-sensitive realization fixes G_Gamma or quantifies uniformly over the admitted family; local Lie-algebra results remain uniform and detailed global-form closure is exported to SYM-9.	AharonySeibergTachikawa2013;Tong2017;MaleyArc
SYM1-T33	Canonical role-normal-form uniqueness	Superseded in v4	Every well-typed gauge object has one locus-indexed normalized role multiset up to permutation and lawful refinement equivalence.	MaleyKernel;MaleyFixedDomain;MaleySYM0
SYM1-T34	Gauge-realization obligation theorem	Superseded in v5	Every complete faithful realization of the frozen target discharges presentation coherence, comparison/transport, physical reduction, matter response, and quantum consistency.	MaleyKernel;MaleyFixedDomain;MaleySYM0
SYM1-T35	Obligation independence and derivability-sensitive minimality	Superseded in v5	Each gauge-realization obligation has an independence model satisfying the other four and is not generated by the declared closure of the remainder.	MaleyKernel;MaleyFixedDomain;MaleySYM0
SYM1-T36	No sixth primitive gauge-realization obligation	Superseded in v5	Every proposed same-scope primitive obligation reduces to the five obligations, target data, downstream dynamics, or derived/representative content.	MaleyKernel;MaleyFixedDomain;MaleySYM0
SYM1-T37	QFT-witness determination	Superseded in v5	Role-neutral QFT packets and proposition-bearing certificates determine the registered extensional witness sets without classifier or role fields.	MaleyKernel;MaleyFixedDomain;MaleySYM0
SYM1-T38	Unique minimal dependency support	Superseded in v5	Every registered certified witness set has a unique minimal support antichain in the finite dependency DAG.	MaleyKernel;MaleyFixedDomain;MaleySYM0
SYM1-T39	Canonical locus-indexed role normal form	Superseded in v5	Minimal support generates a multiplicity-preserving multiset of structured role atoms, unique up to lawful permutation preserving dependency closure.	MaleyKernel;MaleyFixedDomain;MaleySYM0
SYM1-T40	Typed constructor-specific normalization	Superseded in v5	Aggregate, product, descent, cohomology, quotient comparison, boundary restriction, and target extension compute relational witnesses from physical inputs.	MaleyKernel;MaleyFixedDomain;MaleySYM0
SYM1-T41	Target-signature representation theorem	Superseded in v5	Every registered gauge-relevant target variation has a certified well-typed representing term; a same-target omission falsifies adequacy.	MaleyKernel;MaleyFixedDomain;MaleySYM0
SYM1-T42	Certificate-relative sharp-local exhaustion	Superseded in v5	Every certified well-typed candidate generated by the frozen target protocol has a canonical role normal form; atlas membership and hostile-ledger emptiness are not premises.	MaleyKernel;MaleyFixedDomain;MaleySYM0
SYM1-T43	Unique admissible interior interface	Imported theorem	For a fixed standing-instantiated domain, no extensionally competing gate or discrete plural admissible interior survives.	SRC-AASC-UNUS
SYM1-T44	Certified consequence-profile extensionality	Superseded in v6	Two complete valid certificate packets for the same candidate agree on all standing-bearing target, quotient, continuation, response and consistency consequences.	SRC-AASC-UNUS;SRC-AASC-FIXED;SRC-AASC-SYMO
SYM1-T45	Finite dependency-order theorem	Superseded in v6	The ranked consequence dependency relation is acyclic and well founded.	SRC-AASC-FIXED
SYM1-T46	Unique minimal-support theorem	Superseded in v6	Every certified profile has one minimal antichain in each target, channel and locus lineage, covering every certified proposition.	SRC-AASC-FIXED
SYM1-T47	Gauge-profile/interior bridge	Superseded in v6	Inequivalent standing-bearing profiles for one fixed candidate induce inequivalent same-domain interiors.	SRC-AASC-UNUS;SRC-AASC-FIXED
SYM1-T48	Adequate role-normal-form uniqueness	Superseded in v6	Every certified profile has one lawful equivalence class of adequate locus-indexed role normal forms.	SRC-AASC-UNUS;SRC-AASC-FIXED;SRC-AASC-SYMO
SYM1-T49	Intrinsically typed constructor normalization	Superseded in v6	Every declared physical constructor has a typed certificate interface and a normalization rule; no relational role leg is accepted as input.	SRC-QFT-HT;SRC-QFT-BBH;SRC-QFT-BILAL
SYM1-T50	Target-derived protocol coverage	Superseded in v6	Every same-scope gauge-relevant variation of the frozen target is assigned to a parameterized protocol family by the four-locus reduction.	SRC-AASC-FIXED;SRC-AASC-UNUS;SRC-AASC-SYMO
SYM1-T51	Gauge-realization obligation necessity and formal-core independence	Superseded in v6	The five obligations are necessary for the fixed target; their formal cores are independent in a common signature.	SRC-AASC-UNUS;SRC-AASC-R8
SYM1-T52	Certificate-relative sharp-local exhaustion	Superseded in v6	Every certified well-typed candidate generated by the target-derived protocol has one adequate role-normal-form class; an omitted same-target candidate breaks protocol adequacy.	SRC-AASC-UNUS;SRC-AASC-FIXED;SRC-AASC-SYMO
SYM1-T53	Candidate-indexed gauge-semantic interface	Superseded in v8	Each candidate is assigned eight fixed role-neutral QFT propositions; certificate packets decide those propositions rather than selecting effect labels. Superseded as an active closure route by the v8 Role-Occupancy Closure specialization.	SRC-AASC-FIXED;SRC-AASC-SYMO
SYM1-T54	Certified consequence-profile extensionality	Superseded in v8	Two valid complete certificate packets for the same fixed candidate agree on all standing-bearing QFT consequences. Superseded as an active closure route by the v8 Role-Occupancy Closure specialization.	SRC-AASC-UNUS;SRC-AASC-FIXED
SYM1-T55	Ranked dependency well-foundedness and minimal support	Superseded in v8	Strict rank increase in one target/channel/locus lineage yields a well-founded dependency relation and one unique minimal support antichain. Superseded as an active closure route by the v8 Role-Occupancy Closure specialization.	SRC-AASC-FIXED

ID	Title	Status	Statement	Sources
SYM1-T56	Gauge-profile/interior bridge	Superseded in v8	Inequivalent standing-bearing gauge profiles induce extensionally different same-domain admissible interiors. Superseded as an active closure route by the v8 Role-Occupancy Closure specialization.	SRC-AASC-UNUS;SRC-AASC-FIXED
SYM1-T57	Adequate role-normal-form uniqueness	Superseded in v8	Every certified candidate has one lawful-equivalence class of adequate multiplicity-preserving locus-indexed role normal forms. Superseded as an active closure route by the v8 Role-Occupancy Closure specialization.	SRC-AASC-UNUS;SRC-AASC-FIXED;SRC-AASC-SYMO
SYM1-T58	Intrinsically typed constructor normalization	Superseded in v8	Every declared gauge constructor has a source/target typing rule, proposition-bearing semantics, and a normalization interface; no preclassified relational leg is accepted. Superseded L as an active closure route by the v8 Role-Occupancy Closure specialization.	SRC-QFT-HT;SRC-QFT-BBH;SRC-QFT-BILA
SYM1-T59	Target-derived protocol coverage	Superseded in v8	The four fixed-domain variation loci refine into the finite parameterized gauge protocol; an omitted same-target relevant candidate falsifies coverage. Superseded as an active closure route by the v8 Role-Occupancy Closure specialization.	SRC-AASC-FIXED;SRC-AASC-UNUS;SRC-AASC-SYMO
SYM1-T60	Lane A sharp-local realization stress test	Superseded in v8	The exact-dimension quotient, transport, finite-cap closure, finite-cap extension, inductive-union, and cyclicity nodes instantiate distinct typed protocol families without scope promotion. Superseded as an active closure route by the v8 Role-Occupancy Closure specialization.	SRC-AASC-LANEA-COMPANION
SYM1-T61	Certificate-relative sharp-local exhaustion	Superseded in v8	Every certified well-typed candidate generated by the target-derived protocol has one adequate role-normal-form class; atlas membership and hostile-ledger emptiness are not premises. Superseded as an active closure route by the v8 Role-Occupancy Closure specialization.	SRC-AASC-UNUS;SRC-AASC-FIXED;SRC-AASC-SYMO
SYM1-T62	Forced target role-set theorem	Imported and specialized	For each fixed SYM-1 target, the role set is fixed by target architecture before residue inspection.	MaleyRoleOccupancy;MaleySYMO
SYM1-T63	Presentation quotient noncircularity theorem	Locally proved	The residue quotient removes only presentation-equivalent duplication and does not presuppose target-role occupancy.	MaleyRoleOccupancy;MaleySYMO
SYM1-T64	Eligibility formation theorem	Certificate-relative	Every positive target-role eligibility edge is generated by a declared QFT or target-typing certificate.	MaleyRoleOccupancy;MaleySYMO
SYM1-T65	Joint compatibility theorem	Certificate-relative	Complete co-instantiation is tested independently of pointwise eligibility.	MaleyRoleOccupancy;MaleySYMO
SYM1-T66	Live-residue theorem	Locally proved	Live residue is exactly the union of residue appearing in complete compatible preassignments.	MaleyRoleOccupancy;MaleySYMO
SYM1-T67	Standard Model assignment-existence theorem	Certificate-relative local	Every frozen target role set has at least one complete compatible assignment in the controlled registry.	MaleyRoleOccupancy;MaleySYMO
SYM1-T68	Certificate-lawful automorphism theorem	Imported and specialized	Only automorphisms preserving the full invariant tuple identify assignments.	MaleyRoleOccupancy;MaleySYMO
SYM1-T69	Unique lawful gauge-assignment orbit	Superseded in v8	Each frozen SYM-1 target has exactly one complete compatible assignment orbit under certificate-lawful automorphisms. Superseded as an active closure route by the v8 Role-Occupancy Closure specialization.	MaleyRoleOccupancy;MaleySYMO
SYM1-T70	Three-axis noncollapse theorem	Locally proved classifier	Target-role occupancy, ATS anchor/tensor/skin type, and SYM-0 status are independent axes.	MaleyRoleOccupancy;MaleySYMO
SYM1-T71	Conditional SYM-1 sharp-local closure	Superseded in v8	Under complete target role sets, eligibility and compatibility ledgers, closed assignment orbits, and ATS/SYM-0 typing, no untyped same-target sharp-local gauge record survives. Superseded as an active closure route by the v8 Role-Occupancy Closure specialization.	MaleyRoleOccupancy;MaleySYMO
SYM1-T72	Target-role necessity certificate theorem	Locally specialized from imported role-occupancy theorem	Every active target role is fixed before residue inspection and carries an occupant-independent target datum and cost-of-denial certificate.	MaleyRoleOccupancy;MaleyRoleCardinality;MaleyKernel
SYM1-T73	Candidate-protocol independence theorem	Mechanically audited finite theorem	The target-derived protocol generates atlas instances, cross-target views, presentation duplicates, and atlas-external candidates; atlas membership is not a generation premise.	MaleyRoleOccupancy;MaleySYMO
SYM1-T74	Presentation-quotient soundness and completeness	Mechanically audited finite theorem	The frozen presentation quotient collapses all and only presentation-level duplicates and preserves every eligibility and compatibility consequence.	MaleyFixedDomain;MaleyRoleOccupancy;MaleySYMO
SYM1-T75	Complete eligibility-matrix theorem	Superseded by v9 integrated-target theorem package	Every presentation-quotiented residue/target-role pair receives exactly one certified eligible or ineligible disposition; no unresolved pair enters release closure.	MaleyRoleOccupancy;MaleySYMO;GaugeQFTSources
SYM1-T76	Joint-compatibility completeness theorem	Superseded by v9 integrated-target theorem package	Every injective pointwise eligible preassignment is evaluated against the complete target, quotient, response, anomaly, boundary, topology, continuation, and failure predicate.	MaleyRoleOccupancy;MaleyStandaloneSM
SYM1-T77	Live-residue derivation theorem	Superseded by v9 integrated-target theorem package	For every frozen target, live residue is exactly the union of images of all compatible pointwise eligible complete preassignments; every eligible non-live residue has an explicit compatibility-impossibility diagnosis.	MaleyRoleOccupancy
SYM1-T78	Complete assignment enumeration theorem	Superseded by v9 integrated-target theorem package	Every compatible bijection from the target role set onto live residue is enumerated; no intended assignment is assumed as exhaustive.	MaleyRoleOccupancy;MaleyRoleCardinality

ID	Title	Status	Statement	Sources
SYM1-T79	Certificate-lawful automorphism computation theorem	Superseded by v9 integrated-target theorem package	The role/residue permutation group preserving the full certificate invariant tuple is exhaustively computed and its group action on complete assignments is verified.	MaleyRoleOccupancy
SYM1-T80	Unique compatible gauge-role occupancy	Superseded by v9 integrated-target theorem package	For each frozen SYM-1 target, complete compatible assignments are nonempty and form exactly one orbit under the computed certificate-lawful automorphism group.	MaleyRoleOccupancy;MaleyRoleCardinality;MaleyKernel;MaleyFixedDomain
SYM1-T81	Seven-clause role-realization completeness	Superseded by v9 integrated-target theorem package	Candidate, quotient, role-set, eligibility, compatibility, live-residue, and failure completeness hold for all fourteen frozen targets.	MaleyRoleOccupancy;MaleySYM0
SYM1-T82	Target-role saturation with explicit refinement branch	Superseded by v9 integrated-target theorem package	Any proposed additional load-bearing role is governance-equivalent, a same-target refinement counterexample, a physical target update, or non-authoritative support/skin.	MaleyRoleOccupancy;MaleyFixedDomain;MaleyATS
SYM1-T83	Three-axis gauge-status noncollapse	Superseded by v9 integrated-target theorem package	Target-role occupancy, ATS anchor/tensor/skin type, and SYM-0 status are independent classifications and cannot be inferred from one another.	MaleyATS;MaleySYM0;MaleyRoleOccupancy
SYM1-T84	Conditional fixed-target SYM-1 gauge closure	Superseded by v9 integrated-target theorem package	For every frozen SYM-1 target, the role-realization-complete certificate and singleton assignment orbit license the target; no generated same-target candidate remains untyped within the declared SYM-1 scope.	MaleyRoleOccupancy;MaleyRoleCardinality;MaleySYM0
SYM1-T85	Integrated-target fixation	Local theorem	The primary closure theorem is evaluated on one integrated Standard Model gauge target; projection scopes state local QFT assumptions but are not independent closure universes.	MaleySYM0;MaleyArc
SYM1-T86	Occupant-independent integrated role necessity	Local specialization	Eleven genuine architecture and operation roles are forced by target data before candidate residue is inspected; diagnostics, witnesses, obstruction records, and handoffs are excluded from the role set.	MaleyRoleOccupancy;MaleyRoleCardinality
SYM1-T87	Target-signature candidate coverage	Certificate-relative finite theorem	Every candidate expressible in the frozen integrated SYM-1 target signature is emitted by the parameterized protocol; a same-target omitted candidate is an explicit falsifier.	MaleyRoleOccupancy;MaleyFixedDomain
SYM1-T88	Presentation-quotient soundness	Certificate-relative finite theorem	The presentation quotient collapses gauge, basis, transport-natural, BRST-exact, and theta-phase duplicates only after the complete invariant and capability profiles agree.	MaleyRoleOccupancy;MaleyATS
SYM1-T89	Complete integrated eligibility matrix	Finite theorem	Every integrated residue-role pair has a positive or negative certificate derived from occupant-independent capability criteria; the intended assignment is not an input.	MaleyRoleOccupancy
SYM1-T90	Independent integrated compatibility calculation	Finite theorem	All 4,608 pointwise eligible injective preassignments are evaluated under sixteen named compatibility components computed from occupant profiles; exactly one passes.	MaleyRoleOccupancy
SYM1-T91	Derived live residue and complete assignment	Finite theorem	The live residue is the union of compatible preassignment images, and exhaustive enumeration yields one compatible bijection from the eleven integrated roles onto the eleven live occupants.	MaleyRoleOccupancy
SYM1-T92	Nontrivial lawful automorphism test	Finite theorem	A controlled two-chart gauge target has two complete assignments, a four-element certificate-lawful automorphism group, and one assignment orbit.	MaleyRoleOccupancy
SYM1-T93	Integrated Gauge Architecture Occupancy Closure	Conditional endpoint theorem	The integrated certificate passes all seven role-realization-completeness clauses and has one assignment orbit; Role-Occupancy Closure therefore licenses the integrated gauge realization relative to the frozen certificate.	MaleyRoleOccupancy;MaleyRoleCardinality
SYM1-T94	Three-axis gauge classification	Local theorem	Target-role occupancy, ATS anchor/tensor/skin type, and SYM-0 status remain independent for every active atlas record.	MaleyATS;MaleySYM0
SYM1-T95	Architecture/operation rigidity corollary	Local theorem	Proper compact-support gauge operations remain quotient-trivial ATS skin at the proper-bulk locus while coherent presentation, transport, reduction, response, and consistency remain load-bearing.	MaleyGaugeR8;HenneauxTeitelboim1992

C.1 Active composition order

The active theorem spine is



A downstream result may not repair a failed upstream premise within the same certificate. Legacy semantic-profile, minimal-support, and role-normal-form engines are provenance artifacts only and are not cited as active closure proofs.

Source Authority and Claim Coverage

D.1 Authority policy

AASC corpus sources establish the kernel and gauge-architecture dependency spine. Primary or standard gauge-theory sources establish constrained reduction, boundary charges, Faddeev–Popov and BRST constructions, global gauge-fixing obstructions, anomalies, transport functors, holonomy, and line-operator sensitivity. The two source classes perform different evidentiary work.

Source	Title	Type	Authority	Publication role
SRC-AASC-A1	The Gauge Transition Groupoid Theorem	AASC gauge arc	Canonical imported theorem	Transition groupoid
SRC-AASC-A2	Connection as Standing-Preserving Transport Geometry	AASC gauge arc	Canonical imported theorem	Transport comparison
SRC-AASC-A3	Curvature as Transport-Holonomy Obstruction	AASC gauge arc	Canonical imported theorem	Curvature/holonomy residue
SRC-AASC-A4	Representation Admissibility and Charge-Witness Rigidity	AASC gauge arc	Canonical imported theorem	Representation response and charge witnesses
SRC-AASC-A5	Anomaly as Standing-Failure Obstruction	AASC gauge arc	Canonical imported theorem	Gauge-anomaly admissibility
SRC-AASC-AER	Admissibility and the Common Structural Origin of Gauge, Constraint, and Quantum Structure	AASC/constraint theorem	Canonical imported theorem	Constraint surface, characteristic quotient, first-class generators
SRC-AASC-ARC	Symmetry as Constraint, Redescription, and Physical Action	AASC plan	Project architecture	Scope and release gates
SRC-AASC-FIXED	A Fixed-Domain Exhaustion Theorem	AASC theorem	Canonical imported theorem	Factorization, no second gate, role closure
SRC-AASC-KERNEL	Non-Degenerate Construction and the Kernel of Admissibility	AASC theorem	Canonical imported theorem	Kernel necessity, anti-weakening, AMetric boundary
SRC-AASC-LANEA-COMPANION	Companion II: Lane A Sharp-Local Construction	AASC/QFT theorem companion	Canonical downstream realization companion	Worked sharp-local realization chain for quotient, transport, finite-cap closure, inductive-union compatibility, and bounded-base cyclicity
SRC-AASC-PSM1	Lie-Type Gauge Realization Admissibility	AASC specialization	Canonical imported theorem	Lie group/algebra representative specialization
SRC-AASC-PSM2	Product Gauge-Sector Compatibility	AASC specialization	Canonical imported theorem	Product-sector compatibility
SRC-AASC-R8	Gauge Structure from Boundary-Trace Fixation	AASC gauge arc	Canonical imported theorem	Architecture induction
SRC-AASC-SM	The Standalone Standard Model Structure from First Principles	AASC Standard Model	Canonical target import	Fixed SM structural target
SRC-AASC-SYM0	The Symmetry Status Calculus	AASC/QFT classifier	Upstream publication project	Typed records, quotient trichotomy, role and target rules
SRC-AASC-UNUS	Unus Solus Possibilis Est: Corpus Closure and Uniqueness of the Admissible Interior	AASC theorem	Canonical imported apex theorem	Unique same-domain gate, unique admissible interior, no discrete multiplicity, and common-signature independence method
SRC-QFT-ADLER	Axial-Vector Vertex in Spinor Electrodynamics	Perturbative anomaly analysis	Primary source	Historical and technical anomaly derivation
SRC-QFT-AHARONY	Reading between the Lines of Four-Dimensional Gauge Theories	Global form and line operators	Primary source	Global-form preview
SRC-QFT-BARNICH	Local BRST Cohomology in Gauge Theories	BRST cohomology	Primary review	BRST physical cohomology
SRC-QFT-BELL-JACKIEWICZ	A PCAC Puzzle: π^0 to gamma gamma in the Sigma Model	Perturbative anomaly analysis	Primary source	Historical anomaly derivation
SRC-QFT-BILAL	Lectures on Anomalies	Anomaly theory	Primary pedagogical review	Local gauge anomaly and SM cancellation
SRC-QFT-BRST	Renormalization of Gauge Theories	BRST	Primary source	BRST differential
SRC-QFT-DIRAC	Lectures on Quantum Mechanics	Constrained Hamiltonian theory	Primary monograph	First-class constraints and gauge generators
SRC-QFT-FP	Feynman Diagrams for the Yang-Mills Field	Gauge fixing	Primary source	Gauge fixing and ghosts

Source	Title	Type	Authority	Publication role
SRC-QFT-FUJIKAWA	Path-Integral Measure for Gauge-Invariant Fermion Theories	Path-integral anomaly analysis	Primary source	Path-integral measure anomaly derivation
SRC-QFT-GAIOTTO	Generalized Global Symmetries	Generalized symmetry	Primary source	SYM-9 handoff
SRC-QFT-GRIBOV	Quantization of Non-Abelian Gauge Theories	Gauge-fixing obstruction	Primary source	Gribov obstruction
SRC-QFT-HT	Quantization of Gauge Systems	Gauge systems	Standard monograph	Canonical reduction, boundary, BRST
SRC-QFT-KUGO-OJIMA	Local Covariant Operator Formalism of Non-Abelian Gauge Theories and Quark Confinement Problem	Operator BRST quantization	Primary source	BRST charge, state complex and physical-state condition
SRC-QFT-NAKAHARA	Geometry, Topology and Physics	Gauge geometry	Standard monograph	Connection/curvature specialization
SRC-QFT-NOETHER	Invariante Variationsprobleme	QFT/classical field theory	Primary source	Gauge identities and constraints
SRC-QFT-PESKIN	An Introduction to Quantum Field Theory	Quantum field theory	Standard monograph	Gauge fixing, BRST, SM conventions
SRC-QFT-REGGE	Role of Surface Integrals in the Hamiltonian Formulation of General Relativity	Boundary generators	Primary source	Boundary charge exception
SRC-QFT-SCHREIBER-WALDORF	Parallel Transport and Functors	Differential geometry / transport functors	Primary mathematical source	Separate transition data from path transport and characterize principal bundles with connection by smooth transport functors
SRC-QFT-SINGER	Some Remarks on the Gribov Ambiguity	Global gauge fixing	Primary source	No global canonical gauge
SRC-QFT-TONG	Line Operators in the Standard Model	SM global form	Primary source	Global form and line-spectrum handoff
SRC-QFT-WEINBERG	The Quantum Theory of Fields, Volume II	Quantum field theory	Standard monograph	Standard Model gauge structure
SRC-QFT-WILSON	Confinement of Quarks	Wilson loops	Primary source	Holonomy/Wilson observables
SRC-QFT-WITTEN	An SU(2) Anomaly	Global gauge anomaly	Primary source	Global gauge anomaly obstruction
SRC-AASC-ROLE-OCCUPANCY	Role-Occupancy Closure under Admissibility	AASC role-occupancy closure	Canonical imported theorem	Compatible role occupancy
SRC-AASC-ROLE-CARDINALITY	Role Cardinality Closure under Admissibility	AASC role-cardinality precursor	Canonical imported theorem	Cardinality consequence of occupancy

D.2 Claim-link distribution

Source	Atomic claims
SRC-AASC-A1	4
SRC-AASC-A2	8
SRC-AASC-A3	5
SRC-AASC-A4	6
SRC-AASC-A5	8
SRC-AASC-AER	4
SRC-AASC-ARC	3
SRC-AASC-FIXED	4
SRC-AASC-KERNEL	1
SRC-AASC-PSM1	3
SRC-AASC-PSM2	4
SRC-AASC-R8	2
SRC-AASC-ROLE-CARDINALITY	1
SRC-AASC-ROLE-OCCUPANCY	4
SRC-AASC-SM	9
SRC-AASC-SYMO	8
SRC-AASC-UNUS	4
SRC-QFT-AHARONY	7
SRC-QFT-BARNICH	5
SRC-QFT-BILAL	7
SRC-QFT-BRST	4
SRC-QFT-DIRAC	2
SRC-QFT-FP	2
SRC-QFT-GAIOTTO	1
SRC-QFT-GRIBOV	1
SRC-QFT-HT	21
SRC-QFT-KUGO-OJIMA	2
SRC-QFT-NAKAHARA	11
SRC-QFT-REGGE	3
SRC-QFT-SCHREIBER-WALDORF	7
SRC-QFT-SINGER	1

Source	Atomic claims
SRC-QFT-TONG	4
SRC-QFT-WEINBERG	9
SRC-QFT-WILSON	1
SRC-QFT-WITTEN	2

D.3 Frozen corpus integrity

The release contains the canonical AASC and ARC inputs used for the manuscript, together with byte counts and SHA-256 hashes. External primary sources are cited by stable bibliography keys and persistent identifiers. No manuscript paragraph is the only location of a load-bearing status claim.

D.4 Online verification ledger

Citekey	Status	Metadata verified online	Manuscript use
	Verified		
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Adversarial Objections and Final Dispositions

Objection	Reconstruction	Exact boundary
Closure was fragmented into microtargets.	One integrated target now contains the architecture, proper/boundary/topological operations, theta, BRST, and synchronization burdens. Legacy targets are projection scopes.	Full line-operator/generalized-symmetry completion remains SYM-9.
Status records were promoted to roles.	Witnesses, anomaly contributions, charge values, boundary-status tables, Gribov obstruction, and hand-offs remain ATS/SYM-0 overlays.	Eleven genuine architecture/operation roles remain.
Roles were backfitted to occupants.	Every role has an occupant-independent target datum and cost-of-denial certificate.	A same-target missing burden is a role-set-refinement counterexample.
The candidate universe was atlas-derived.	A frozen target signature and protocol families generate atlas views, composite packages, diagnostics, and atlas-external candidates.	Missing same-target generated content reopens candidate completeness.
Presentation quotient was trivial.	Five nontrivial gauge, transport, weak-basis, BRST, and theta presentation pairs are collapsed only after invariant agreement.	Tensor-, boundary-, topology-, anomaly-, and compatibility-bearing differences remain.
Eligibility predetermined assignment.	Every residue/role pair is classified by required capability; multiple roles have multiple eligible residues.	Desired assignment is not an eligibility input.
Compatibility defaulted to acceptance.	Sixteen named components are computed from candidate profiles for all 4,608 integrated preassignments.	New same-target co-instantiation data reopen compatibility completeness.

Objection	Reconstruction	Exact boundary
Live residue was selected.	It is the union of all compatible preassignment images.	Pointwise eligible alternatives absent from that union carry failure diagnoses.
Automorphism quotient was decorative.	A real two-chart target yields two assignments, a four-element group, and one orbit.	The integrated target remains rigid; no physical distinctions are quotient-collapsed.
Completeness was field population.	Each clause is tied to an independently generated registry and explicit same-target falsifier.	Completeness remains relative to the frozen target signature.
Formal package over-claimed.	The Lean source is an interface specification; the executable Python model checker performs the finite enumeration.	Continuum QFT inputs remain imported theorem certificates.

E.1 Adversarial finite test suite

ID	Adversarial test	Result	Evidence
HT9-01	Integrated rather than fragmented primary target	PASS	One primary 11-role target contains architecture, proper/boundary/topological operations, theta, BRST, and synchronization.
HT9-02	Status records not promoted to target roles	PASS	Witnesses, obstruction records, and handoffs remain ATS/SYM-0 overlays.
HT9-03	Atlas-external generation	PASS	Target signature generated a candidate not present in the atlas.
HT9-04	Multiple presentation equivalence classes	PASS	Gauge, basis, BRST representative, transport, and phase-convention duplicates are tested.
HT9-05	Complete eligibility matrix	PASS	752 candidate-role pairs classified.
HT9-06	Nontrivial pointwise competition	PASS	At least eight integrated roles have multiple eligible residues.
HT9-07	All eligible preassignments enumerated	PASS	All products of positive role-incidence choices are enumerated subject to injectivity.
HT9-08	Compatibility not pointwise factorized	PASS	Pointwise eligible alternatives fail named transport, quotient, boundary, topology, theta, BRST, anomaly, or synchronization components.
HT9-09	Live residue derived	PASS	Live residue is the union of compatible preassignment images.
HT9-10	Non-live eligible diagnoses	PASS	Individually eligible but compatibility-impossible residues carry failure components.
HT9-11	Complete assignments enumerated	PASS	All compatible bijections onto computed live residue are enumerated.
HT9-12	Nontrivial lawful automorphism	PASS	Two interchangeable chart roles/residues produce two assignments in one S2 orbit.
HT9-13	Integrated automorphism does not over-quotient	PASS	All integrated role/residue invariant signatures are rigidly distinguished.
HT9-14	Seven completeness clauses	PASS	Every active closure target passes all seven role-realization-completeness clauses relative to the frozen signature.
HT9-15	Same-target role refinement remains falsifier	PASS	A missing same-target role reopens the theorem.
HT9-16	Proper operation does not demote architecture	PASS	Proper operation is separately typed from presentation/-transport/reduction/response/consistency roles.

ID	Adversarial test	Result	Evidence
HT9-17	Projection targets not used as microtarget closure proof	PASS	Only integrated and controlled automorphism targets are active occupancy universes; legacy scopes are projections.
HT9-18	Role-occupancy endpoint	PASS	Integrated certificate has one lawful assignment orbit and passes completeness.

Formal Specification Boundary

F.1 Finite objects

The active companion specifies finite role sets, candidate residue, presentation equivalence, eligibility, joint compatibility, preassignments, live residue, complete assignments, lawful automorphisms, assignment orbits, and role-realization completeness.

F.2 Imported interfaces

Kernel necessity, fixed-domain gate discipline, ATS, SYM-0, Role-Occupancy Closure, and continuum-QFT theorems are explicit imported interfaces. The finite occupancy calculation begins only after their target-indexed consequences have been entered as certificate facts.

F.3 Executable verification

The Python generator is authoritative for the finite calculation. It derives the target role set, candidate universe, presentation quotient, complete eligibility matrix, all preassignments, compatibility values, live residue, assignments, automorphisms, orbits, and completeness certificates. The audit regenerates and checks these objects independently. The Lean files express the corresponding finite types and theorem interfaces; Lean is not installed in the release environment, so the project does not claim compiled Lean verification.

F.4 Prohibited shortcuts

The active closure engine contains no stored intended assignment used to generate eligibility, no compatibility coordinate initialized to acceptance, no atlas-membership premise, no cardinality-only sufficiency rule, no empty-rival-ledger premise, and no vacuous truth constant standing in for adequacy.

Exact Nonclaims

ID	Blocked inference	Scope
NC-01		
NC-02		
NC-03		
NC-04		
NC-05		
NC-06		
NC-07		
NC-08		
NC-09		
NC-10		

G.1 Interpretive clarifications

“Skin” does not mean unreal, useless, dispensable from calculation, or empirically irrelevant. It means that the declared variation changes only a representative after target and quotient-level standing content are fixed. “Derived witness” does not mean approximate. “Anchor” is not a demotion. “Target update” is not a rejection; it is a scope classification.

No result establishes that every gauge theory has the Standard Model local group, that the Standard Model is unique under arbitrary premises, that numerical couplings are forced, that every global form has been selected, or that generalized symmetries and ultraviolet completions have been exhausted.

Evidence and Role-Occupancy Certificate Index

Every active atlas record retains its synchronized evidence directory. The v9 integrated role-occupancy layer adds occupant-independent role-necessity certificates, presentation-equivalence certificates, complete positive and negative eligibility dispositions, compatibility evaluations for every eligible injective assignment, non-live diagnoses, complete assignment records, computed automorphism groups, orbit registries, and seven-part role-realization-completeness certificates.

Record	Target	Quotient	Quantum	Role/status
SYM-01-GOVERNANCE-Q UOTIENT-v2				
SYM-01-QUOTIENT-FID ELITY-BRIDGE-v2				
SYM-01-CHARGE-NORMA LIZATION-v1				
SYM-01-CHARGE-WITNE SS-v1				
SYM-01-CONNECTION-R OLE-v1				
SYM-01-CURVATURE-RO LE-v1				
SYM-01-GAUGE-ARCH-v 1				
SYM-01-HOLONOMY-CLA SS-v1				
SYM-01-LIE-SPECIALI ZATION-v1				
SYM-01-PRODUCT-COMP ATIBILITY-v1				
SYM-01-REPRESENTATI ON-RESPONSE-v1				
SYM-01-SM-ASSEMBLY- CERT-v1				
SYM-01-SM-LOCAL-PRO DUCT-v1				
SYM-01-TRANSITION-G ROUPOID-v1				
SYM-01-GAUGE-NATURA L-EQUIVALENCE-v2				
SYM-01-THIN-PATH-GR OUPOID-v2				
SYM-01-TRANSPORT-FU NCTOR-v2				
SYM-01-CLASSICAL-GA UGE-QUOTIENT-v2				
SYM-01-CONNECTION-D YNAMICAL-VARIATION- v2				
SYM-01-CONNECTION-L OCAL-COORDINATES-v2				
SYM-01-CONNECTION-O RBIT-CLASS-v2				
SYM-01-CONNECTION-V ERTICAL-VARIATION-v 2				
SYM-01-CONSTRAINT-S URFACE-v1				
SYM-01-CURVATURE-FO RM-v1				
SYM-01-GAUGE-LOCAL- PROPER-v1				
SYM-01-GAUSS-GENERA TOR-PROPER-v1				
SYM-01-REPRESENTATI ON-BASIS-v1				
SYM-01-GAUGE-BOUND ARY-ACTION-v1				

Record	Target	Quotient	Quantum	Role/status
SYM-01-GAUGE-BOUNDARY-RULE-v1				
SYM-01-SURFACE-CHARGE-v1				
SYM-01-GAUGE-LARGE-EQUIVARIANCE-v2				
SYM-01-GAUGE-LARGE-QUOTIENT-v1				
SYM-01-GAUGE-LARGE-SECTOR-ACTION-RULE-v2				
SYM-01-THETA-SECTOR-DATUM-v2				
SYM-01-TOPOLOGY-ANCHOR-v1				
SYM-01-TOPOLOGY-UPDATE-v1				
SYM-01-BRST-DIFFERENTIAL-v1				
SYM-01-BRST-REALIZATION-ARCH-v2				
SYM-01-FP-GHOSTS-v1				
SYM-01-GAUGE-FIXING-v1				
SYM-01-GHOST-NUMBER-v1				
SYM-01-GRIBOV-OBSTRUCTION-v1				
SYM-01-BRST-STATE-COHOMOLOGY-v2				
SYM-01-BRST-LOCAL-COHOMOLOGY-v2				
SYM-01-BRST-RELATIVE-COHOMOLOGY-v2				
SYM-01-AGGREGATE-REPLY-LEDGER-v2				
SYM-01-ANOMALY-NEUTRALITY-v1				
SYM-01-INFLOW-UPDATE-v1				
SYM-01-LOCAL-GAUGE-ANOMALY-v1				
SYM-01-REP-ANOMALY-CONTRIBUTION-v2				
SYM-01-SM-ANOMALY-CANCELLATION-v1				
SYM-01-SU2-GLOBAL-ANOMALY-v1				
SYM-01-GLOBAL-FORM-PLACEHOLDER-v1				
SYM-01-WILSON-LOOP-v1				
SYM-01-COUPLING-NONCLAIM-v1				
SYM-01-EW-VACUUM-HANDOFF-v1				
SYM-01-LINE-OPERATOR-HANDOFF-v1				

H.1 Acceptance rule

A target enters the sufficiency direction of Role-Occupancy Closure only when candidate, quotient, role-set, eligibility, compatibility, live-residue, and failure completeness all pass; at least one complete assignment exists; and the computed assignment quotient has exactly one orbit. A blocked, unresolved, refuted, superseded, or target-mismatched certificate fails closed.

H.2 Counterexample protocol

A counterexample states the fixed target; proposed missing role or candidate; acted-upon object; transformation or relation; quotient; boundary, topology, and global-form data; candidate-role eligibility; compatibility consequences; surviving live residue or assignment; claimed lawful-equivalence status; source assumptions; and the theorem it contradicts. A genuine same-target counterexample triggers a versioned revision rather than protective relabeling.

Complete Integrated Role-Occupancy Matrices

I.1 Full eligibility matrix

The table records every residue/role pair for the integrated target and controlled automorphism target. Positive and negative dispositions are both certificate objects.

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

Residue	Role	Result	Certified criterion result
RES9-SYM-01-TOPOLOGY-UPDATE-v1	R9-BRST-REALIZATION	Ineligible	Certified candidate facts do not include required capability brst_realization_architecture.
RES9-SYM-01-TOPOLOGY-UPDATE-v1	R9-SYNCHRONIZATION	Ineligible	Certified candidate facts do not include required capability synchronization_architecture.
RES9-SYM-01-TRANSITION-GROUPOID-v1	R9-PRESENTATION	Eligible	Certified candidate facts include required capability presentation_architecture; the edge is formed by target-internal typing fixed before assignment enumeration.
RES9-SYM-01-TRANSITION-GROUPOID-v1	R9-COMPARISON	Ineligible	Certified candidate facts do not include required capability comparison_architecture.
RES9-SYM-01-TRANSITION-GROUPOID-v1	R9-REDUCTION	Ineligible	Certified candidate facts do not include required capability reduction_architecture.
RES9-SYM-01-TRANSITION-GROUPOID-v1	R9-RESPONSE	Ineligible	Certified candidate facts do not include required capability response_architecture.
RES9-SYM-01-TRANSITION-GROUPOID-v1	R9-CONSISTENCY	Ineligible	Certified candidate facts do not include required capability consistency_architecture.
RES9-SYM-01-TRANSITION-GROUPOID-v1	R9-PROPER-OPERATION	Ineligible	Certified candidate facts do not include required capability proper_operation_class.
RES9-SYM-01-TRANSITION-GROUPOID-v1	R9-BOUNDARY-ACTION	Ineligible	Certified candidate facts do not include required capability boundary_action_architecture.
RES9-SYM-01-TRANSITION-GROUPOID-v1	R9-GLOBALFORM-TOPOLOGY	Ineligible	Certified candidate facts do not include required capability global-form_topology_architecture.
RES9-SYM-01-TRANSITION-GROUPOID-v1	R9-THETA-REALIZATION	Ineligible	Certified candidate facts do not include required capability theta_realization_architecture.
RES9-SYM-01-TRANSITION-GROUPOID-v1	R9-BRST-REALIZATION	Ineligible	Certified candidate facts do not include required capability brst_realization_architecture.
RES9-SYM-01-TRANSITION-GROUPOID-v1	R9-SYNCHRONIZATION	Ineligible	Certified candidate facts do not include required capability synchronization_architecture.
RES9-SYM-01-TRANSPORT-FUNCTOR-v2	R9-PRESENTATION	Ineligible	Certified candidate facts do not include required capability presentation_architecture.
RES9-SYM-01-TRANSPORT-FUNCTOR-v2	R9-COMPARISON	Eligible	Certified candidate facts include required capability comparison_architecture; the edge is formed by target-internal typing fixed before assignment enumeration.
RES9-SYM-01-TRANSPORT-FUNCTOR-v2	R9-REDUCTION	Ineligible	Certified candidate facts do not include required capability reduction_architecture.
RES9-SYM-01-TRANSPORT-FUNCTOR-v2	R9-RESPONSE	Ineligible	Certified candidate facts do not include required capability response_architecture.
RES9-SYM-01-TRANSPORT-FUNCTOR-v2	R9-CONSISTENCY	Ineligible	Certified candidate facts do not include required capability consistency_architecture.
RES9-SYM-01-TRANSPORT-FUNCTOR-v2	R9-PROPER-OPERATION	Ineligible	Certified candidate facts do not include required capability proper_operation_class.
RES9-SYM-01-TRANSPORT-FUNCTOR-v2	R9-BOUNDARY-ACTION	Ineligible	Certified candidate facts do not include required capability boundary_action_architecture.
RES9-SYM-01-TRANSPORT-FUNCTOR-v2	R9-GLOBALFORM-TOPOLOGY	Ineligible	Certified candidate facts do not include required capability global-form_topology_architecture.
RES9-SYM-01-TRANSPORT-FUNCTOR-v2	R9-THETA-REALIZATION	Ineligible	Certified candidate facts do not include required capability theta_realization_architecture.
RES9-SYM-01-TRANSPORT-FUNCTOR-v2	R9-BRST-REALIZATION	Ineligible	Certified candidate facts do not include required capability brst_realization_architecture.
RES9-SYM-01-TRANSPORT-FUNCTOR-v2	R9-SYNCHRONIZATION	Ineligible	Certified candidate facts do not include required capability synchronization_architecture.
RES9-SYM-01-WILSON-LOOP-v1	R9-PRESENTATION	Ineligible	Certified candidate facts do not include required capability presentation_architecture.
RES9-SYM-01-WILSON-LOOP-v1	R9-COMPARISON	Ineligible	Certified candidate facts do not include required capability comparison_architecture.
RES9-SYM-01-WILSON-LOOP-v1	R9-REDUCTION	Ineligible	Certified candidate facts do not include required capability reduction_architecture.
RES9-SYM-01-WILSON-LOOP-v1	R9-RESPONSE	Ineligible	Certified candidate facts do not include required capability response_architecture.
RES9-SYM-01-WILSON-LOOP-v1	R9-CONSISTENCY	Ineligible	Certified candidate facts do not include required capability consistency_architecture.
RES9-SYM-01-WILSON-LOOP-v1	R9-PROPER-OPERATION	Ineligible	Certified candidate facts do not include required capability proper_operation_class.
RES9-SYM-01-WILSON-LOOP-v1	R9-BOUNDARY-ACTION	Ineligible	Certified candidate facts do not include required capability boundary_action_architecture.
RES9-SYM-01-WILSON-LOOP-v1	R9-GLOBALFORM-TOPOLOGY	Ineligible	Certified candidate facts do not include required capability global-form_topology_architecture.
RES9-SYM-01-WILSON-LOOP-v1	R9-THETA-REALIZATION	Ineligible	Certified candidate facts do not include required capability theta_realization_architecture.
RES9-SYM-01-WILSON-LOOP-v1	R9-BRST-REALIZATION	Ineligible	Certified candidate facts do not include required capability brst_realization_architecture.
RES9-SYM-01-WILSON-LOOP-v1	R9-SYNCHRONIZATION	Ineligible	Certified candidate facts do not include required capability synchronization_architecture.
RES9-V9-PKG-BOUNDARY	R9-PRESENTATION	Ineligible	Certified candidate facts do not include required capability presentation_architecture.
RES9-V9-PKG-BOUNDARY	R9-COMPARISON	Ineligible	Certified candidate facts do not include required capability comparison_architecture.
RES9-V9-PKG-BOUNDARY	R9-REDUCTION	Ineligible	Certified candidate facts do not include required capability reduction_architecture.
RES9-V9-PKG-BOUNDARY	R9-RESPONSE	Ineligible	Certified candidate facts do not include required capability response_architecture.
RES9-V9-PKG-BOUNDARY	R9-CONSISTENCY	Ineligible	Certified candidate facts do not include required capability consistency_architecture.
RES9-V9-PKG-BOUNDARY	R9-PROPER-OPERATION	Ineligible	Certified candidate facts do not include required capability proper_operation_class.
RES9-V9-PKG-BOUNDARY	R9-BOUNDARY-ACTION	Eligible	Certified candidate facts include required capability boundary_action_architecture; the edge is formed by target-internal typing fixed before assignment enumeration.
RES9-V9-PKG-BOUNDARY	R9-GLOBALFORM-TOPOLOGY	Ineligible	Certified candidate facts do not include required capability global-form_topology_architecture.
RES9-V9-PKG-BOUNDARY	R9-THETA-REALIZATION	Ineligible	Certified candidate facts do not include required capability theta_realization_architecture.
RES9-V9-PKG-BOUNDARY	R9-BRST-REALIZATION	Ineligible	Certified candidate facts do not include required capability brst_realization_architecture.
RES9-V9-PKG-BOUNDARY	R9-SYNCHRONIZATION	Ineligible	Certified candidate facts do not include required capability synchronization_architecture.
RES9-V9-PKG-BOUNDARY-NONINTEGRABLE	R9-PRESENTATION	Ineligible	Certified candidate facts do not include required capability presentation_architecture.
RES9-V9-PKG-BOUNDARY-NONINTEGRABLE	R9-COMPARISON	Ineligible	Certified candidate facts do not include required capability comparison_architecture.
RES9-V9-PKG-BOUNDARY-NONINTEGRABLE	R9-REDUCTION	Ineligible	Certified candidate facts do not include required capability reduction_architecture.
RES9-V9-PKG-BOUNDARY-NONINTEGRABLE	R9-RESPONSE	Ineligible	Certified candidate facts do not include required capability response_architecture.
RES9-V9-PKG-BOUNDARY-NONINTEGRABLE	R9-CONSISTENCY	Ineligible	Certified candidate facts do not include required capability consistency_architecture.
RES9-V9-PKG-BOUNDARY-NONINTEGRABLE	R9-PROPER-OPERATION	Ineligible	Certified candidate facts do not include required capability proper_operation_class.

Residue	Role	Result	Certified criterion result
RES9-V9-PKG-BOUNDARY-NONINTEGRABLE	R9-BOUNDARY-ACTION	Eligible	Certified candidate facts include required capability boundary_action_architecture; the edge is formed by target-internal typing fixed before assignment enumeration.
RES9-V9-PKG-BOUNDARY-NONINTEGRABLE	R9-GLOBALFORM-TOPOLOGY	Ineligible	Certified candidate facts do not include required capability global-form_topology_architecture.
RES9-V9-PKG-BOUNDARY-NONINTEGRABLE	R9-THETA-REALIZATION	Ineligible	Certified candidate facts do not include required capability theta_realization_architecture.
RES9-V9-PKG-BOUNDARY-NONINTEGRABLE	R9-BRST-REALIZATION	Ineligible	Certified candidate facts do not include required capability brst_realization_architecture.
RES9-V9-PKG-BOUNDARY-NONINTEGRABLE	R9-SYNCHRONIZATION	Ineligible	Certified candidate facts do not include required capability synchronization_architecture.
RES9-V9-PKG-BRST	R9-PRESENTATION	Ineligible	Certified candidate facts do not include required capability presentation_architecture.
RES9-V9-PKG-BRST	R9-COMPARISON	Ineligible	Certified candidate facts do not include required capability comparison_architecture.
RES9-V9-PKG-BRST	R9-REDUCTION	Ineligible	Certified candidate facts do not include required capability reduction_architecture.
RES9-V9-PKG-BRST	R9-RESPONSE	Ineligible	Certified candidate facts do not include required capability response_architecture.
RES9-V9-PKG-BRST	R9-CONSISTENCY	Ineligible	Certified candidate facts do not include required capability consistency_architecture.
RES9-V9-PKG-BRST	R9-PROPER-OPERATION	Ineligible	Certified candidate facts do not include required capability proper_operation_class.
RES9-V9-PKG-BRST	R9-BOUNDARY-ACTION	Ineligible	Certified candidate facts do not include required capability boundary_action_architecture.
RES9-V9-PKG-BRST	R9-GLOBALFORM-TOPOLOGY	Ineligible	Certified candidate facts do not include required capability global-form_topology_architecture.
RES9-V9-PKG-BRST	R9-THETA-REALIZATION	Ineligible	Certified candidate facts do not include required capability theta_realization_architecture.
RES9-V9-PKG-BRST	R9-BRST-REALIZATION	Eligible	Certified candidate facts include required capability brst_realization_architecture; the edge is formed by target-internal typing fixed before assignment enumerat...
RES9-V9-PKG-BRST	R9-SYNCHRONIZATION	Ineligible	Certified candidate facts do not include required capability synchronization_architecture.
RES9-V9-PKG-CONSISTENCY	R9-PRESENTATION	Ineligible	Certified candidate facts do not include required capability presentation_architecture.
RES9-V9-PKG-CONSISTENCY	R9-COMPARISON	Ineligible	Certified candidate facts do not include required capability comparison_architecture.
RES9-V9-PKG-CONSISTENCY	R9-REDUCTION	Ineligible	Certified candidate facts do not include required capability reduction_architecture.
RES9-V9-PKG-CONSISTENCY	R9-RESPONSE	Ineligible	Certified candidate facts do not include required capability response_architecture.
RES9-V9-PKG-CONSISTENCY	R9-CONSISTENCY	Eligible	Certified candidate facts include required capability consistency_architecture; the edge is formed by target-internal typing fixed before assignment enumeration.
RES9-V9-PKG-CONSISTENCY	R9-PROPER-OPERATION	Ineligible	Certified candidate facts do not include required capability proper_operation_class.
RES9-V9-PKG-CONSISTENCY	R9-BOUNDARY-ACTION	Ineligible	Certified candidate facts do not include required capability boundary_action_architecture.
RES9-V9-PKG-CONSISTENCY	R9-GLOBALFORM-TOPOLOGY	Ineligible	Certified candidate facts do not include required capability global-form_topology_architecture.
RES9-V9-PKG-CONSISTENCY	R9-THETA-REALIZATION	Ineligible	Certified candidate facts do not include required capability theta_realization_architecture.
RES9-V9-PKG-CONSISTENCY	R9-BRST-REALIZATION	Ineligible	Certified candidate facts do not include required capability brst_realization_architecture.
RES9-V9-PKG-CONSISTENCY	R9-SYNCHRONIZATION	Ineligible	Certified candidate facts do not include required capability synchronization_architecture.
RES9-V9-PKG-GLOBALFORM-TOPOLOGY	R9-PRESENTATION	Ineligible	Certified candidate facts do not include required capability presentation_architecture.
RES9-V9-PKG-GLOBALFORM-TOPOLOGY	R9-COMPARISON	Ineligible	Certified candidate facts do not include required capability comparison_architecture.
RES9-V9-PKG-GLOBALFORM-TOPOLOGY	R9-REDUCTION	Ineligible	Certified candidate facts do not include required capability reduction_architecture.
RES9-V9-PKG-GLOBALFORM-TOPOLOGY	R9-RESPONSE	Ineligible	Certified candidate facts do not include required capability response_architecture.
RES9-V9-PKG-GLOBALFORM-TOPOLOGY	R9-CONSISTENCY	Ineligible	Certified candidate facts do not include required capability consistency_architecture.
RES9-V9-PKG-GLOBALFORM-TOPOLOGY	R9-PROPER-OPERATION	Ineligible	Certified candidate facts do not include required capability proper_operation_class.
RES9-V9-PKG-GLOBALFORM-TOPOLOGY	R9-BOUNDARY-ACTION	Ineligible	Certified candidate facts do not include required capability boundary_action_architecture.
RES9-V9-PKG-GLOBALFORM-TOPOLOGY	R9-GLOBALFORM-TOPOLOGY	Eligible	Certified candidate facts include required capability globalform_topology_architecture; the edge is formed by target-internal typing fixed before assignment enume...
RES9-V9-PKG-GLOBALFORM-TOPOLOGY	R9-THETA-REALIZATION	Ineligible	Certified candidate facts do not include required capability theta_realization_architecture.
RES9-V9-PKG-GLOBALFORM-TOPOLOGY	R9-BRST-REALIZATION	Ineligible	Certified candidate facts do not include required capability brst_realization_architecture.
RES9-V9-PKG-GLOBALFORM-TOPOLOGY	R9-SYNCHRONIZATION	Ineligible	Certified candidate facts do not include required capability synchronization_architecture.
RES9-V9-PKG-PRESENTATION	R9-PRESENTATION	Eligible	Certified candidate facts include required capability presentation_architecture; the edge is formed by target-internal typing fixed before assignment enumeration.
RES9-V9-PKG-PRESENTATION	R9-COMPARISON	Ineligible	Certified candidate facts do not include required capability comparison_architecture.
RES9-V9-PKG-PRESENTATION	R9-REDUCTION	Ineligible	Certified candidate facts do not include required capability reduction_architecture.
RES9-V9-PKG-PRESENTATION	R9-RESPONSE	Ineligible	Certified candidate facts do not include required capability response_architecture.
RES9-V9-PKG-PRESENTATION	R9-CONSISTENCY	Ineligible	Certified candidate facts do not include required capability consistency_architecture.
RES9-V9-PKG-PRESENTATION	R9-PROPER-OPERATION	Ineligible	Certified candidate facts do not include required capability proper_operation_class.
RES9-V9-PKG-PRESENTATION	R9-BOUNDARY-ACTION	Ineligible	Certified candidate facts do not include required capability boundary_action_architecture.
RES9-V9-PKG-PRESENTATION	R9-GLOBALFORM-TOPOLOGY	Ineligible	Certified candidate facts do not include required capability global-form_topology_architecture.
RES9-V9-PKG-PRESENTATION	R9-THETA-REALIZATION	Ineligible	Certified candidate facts do not include required capability theta_realization_architecture.
RES9-V9-PKG-PRESENTATION	R9-BRST-REALIZATION	Ineligible	Certified candidate facts do not include required capability brst_realization_architecture.
RES9-V9-PKG-PRESENTATION	R9-SYNCHRONIZATION	Ineligible	Certified candidate facts do not include required capability synchronization_architecture.
RES9-V9-PKG-REDUCTION	R9-PRESENTATION	Ineligible	Certified candidate facts do not include required capability presentation_architecture.
RES9-V9-PKG-REDUCTION	R9-COMPARISON	Ineligible	Certified candidate facts do not include required capability comparison_architecture.

[illegible]

Residue	Role	Result	Certified criterion result
RES9-V9-TARGET-DERIVED-WARD	R9-SYNCHRONIZATION	Ineligible	Certified candidate facts do not include required capability synchronization_architecture.
RES9-CHART-A	R9-CHART-A	Eligible	Certified candidate facts include required capability chart_presentation; the edge is formed by target-internal typing fixed before assignment enumeration.
RES9-CHART-A	R9-CHART-B	Eligible	Certified candidate facts include required capability chart_presentation; the edge is formed by target-internal typing fixed before assignment enumeration.
RES9-CHART-B	R9-CHART-A	Eligible	Certified candidate facts include required capability chart_presentation; the edge is formed by target-internal typing fixed before assignment enumeration.
RES9-CHART-B	R9-CHART-B	Eligible	Certified candidate facts include required capability chart_presentation; the edge is formed by target-internal typing fixed before assignment enumeration.

I.2 Compatibility and live residue

The complete 4,608-row preassignment registry is supplied as JSON/CSV. The manuscript reports the exhaustive failure counts and the computed eligible/live sets.

Integrated compatibility object	Count
Pointwise eligible injective preassignments	4608
Jointly compatible preassignments	1
Pointwise eligible residue	24
Live residue	11
Complete assignments	1

Failed compatibility component	Rejected
quotient_crosscheck	4224
theta_physicality	3840
globalform_theta_crosscheck	3840
aggregate_anomaly_consistency	3456
response_anomaly_crosscheck	3456
boundary_topology_crosscheck	3456
physical_reduction	3072
presentation_coherence	2304
transport_completeness	2304
matter_response	2304
proper_operation_scope	2304
globalform_topology	2304
brst_realization	2304
cross_realization_synchronization	2304
boundary_integrability	2304

Class	Residue identifiers
D-SYM1-SM-ASSEMBLY: eligible	RES9-SYM-01-ANOMALY-NEUTRALITY-v1; RES9-SYM-01-BRST-REALIZATION-ARCH-v2; RES9-SYM-01-CHARGE-WITNESS-v1; RES9-SYM-01-CLASSICAL-GAUGE-QUOTIENT-v2; RES9-SYM-01-CONNECTION-ROLE-v1; RES9-SYM-01-CONNECTION-VERTICAL-VARIATION-v2; RES9-SYM-01-GAUGE-LARGE-SECTOR-ACTION-RULE-v2; RES9-SYM-01-GAUGE-LOCAL-PROPER-v1; RES9-SYM-01-GLOBAL-FORM-PLACEHOLDER-v1; RES9-SYM-01-GOVERNANCE-QUOTIENT-v2; RES9-SYM-01-REPRESENTATION-RESPONSE-v1; RES9-SYM-01-SM-ASSEMBLY-CERT-v1; RES9-SYM-01-THETA-SECTOR-DATUM-v2; RES9-SYM-01-TRANSITION-GROUPOID-v1; RES9-SYM-01-TRANSPORT-FUNCTOR-v2; RES9-V9-PKG-BOUNDARY; RES9-V9-PKG-BOUNDARY-NONINTEGRABLE; RES9-V9-PKG-BRST; RES9-V9-PKG-CONSISTENCY; RES9-V9-PKG-GLOBALFORM-TOPOLOGY; RES9-V9-PKG-PRESENTATION; RES9-V9-PKG-REDUCTION; RES9-V9-PKG-SYNCHRONIZATION; RES9-V9-PKG-THETA
D-SYM1-SM-ASSEMBLY: live	RES9-SYM-01-GAUGE-LOCAL-PROPER-v1; RES9-SYM-01-REPRESENTATION-RESPONSE-v1; RES9-SYM-01-TRANSPORT-FUNCTOR-v2; RES9-V9-PKG-BOUNDARY; RES9-V9-PKG-BRST; RES9-V9-PKG-CONSISTENCY; RES9-V9-PKG-GLOBALFORM-TOPOLOGY; RES9-V9-PKG-PRESENTATION; RES9-V9-PKG-REDUCTION; RES9-V9-PKG-SYNCHRONIZATION; RES9-V9-PKG-THETA
D-SYM1-TWO-CHART-PRESENTATION: eligible	RES9-CHART-A; RES9-CHART-B
D-SYM1-TWO-CHART-PRESENTATION: live	RES9-CHART-A; RES9-CHART-B

Target	Non-live residue	Failed components
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-ANOMALY-NEUTRALITY-v1	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-BRST-REALIZATION-ARCH-v2	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-CHARGE-WITNESS-v1	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-CLASSICAL-GAUGE-QUOTIENT-v2	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-CONNECTION-ROLE-v1	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-CONNECTION-VERITICAL-VARIATION-v2	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-GAUGE-LARGE-SECTOR-ACTION-RULE-v2	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-GLOBAL-FORM-PLACEHOLDER-v1	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness

Target	Non-live residue	Failed components
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-GOVERNANCE-QUOTIENT-v2	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-SM-ASSEMBLY-CERT-v1	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-THETA-SECTOR-DATUM-v2	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-SYM-01-TRANSITION-GROUP-UP-IDENT-v1	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness
D-SYM1-SM-ASSEMBLY	RES9-V9-PKG-BOUNDARY-NONINTEGRABLE	aggregate_anomaly_consistency; boundary_integrability; boundary_topology_crosscheck; brst_realization; cross_realization_synchronization; globalform_theta_crosscheck; globalform_topology; matter_response; physical_reduction; presentation_coherence; proper_operation_scope; quotient_crosscheck; response_anomaly_crosscheck; theta_physicality; transport_completeness

I.3 Assignments, lawful automorphisms, and completeness

Target role	Live occupant
R9-PRESENTATION	RES9-V9-PKG-PRESENTATION
R9-COMPARISON	RES9-SYM-01-TRANSPORT-FUNCTOR-v2
R9-REDUCTION	RES9-V9-PKG-REDUCTION
R9-RESPONSE	RES9-SYM-01-REPRESENTATION-RESPONSE-v1
R9-CONSISTENCY	RES9-V9-PKG-CONSISTENCY
R9-PROPER-OPERATION	RES9-SYM-01-GAUGE-LOCAL-PROPER-v1
R9-BOUNDARY-ACTION	RES9-V9-PKG-BOUNDARY
R9-GLOBALFORM-TOPOLOGY	RES9-V9-PKG-GLOBALFORM-TOPOLOGY
R9-THETA-REALIZATION	RES9-V9-PKG-THETA
R9-BRST-REALIZATION	RES9-V9-PKG-BRST
R9-SYNCHRONIZATION	RES9-V9-PKG-SYNCHRONIZATION

Target	Assign.	$ \Gamma $	Orbits	Closed	Orbit characterization
D-SYM1-SM-ASSEMBLY	1	1	1	yes	Rigid integrated assignment.
D-SYM1-TWO-CHART-PRESENTATION	2	4	1	yes	Two chart assignments identified by the four-element lawful incidence group.

Target	Cand.	Quot.	Roles	Elig.	Compat.	Live	Fail	Endpoint
D-SYM1-SM-ASSEMBLY	P	P	P	P	P	P	P	Closed (relative)
D-SYM1-TWO-CHART-PRESENTATION	P	P	P	P	P	P	P	Closed (relative)

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